

Synaptic mechanisms modulate the spatiotemporal dynamics of striatal direct pathway neurons and motor output

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John J Marshall , Jian Xu, Nai-Hsing Yeh, Seongsik Yun, Toshihiro Nomura, John N Armstrong, Jones G Parker , Anis Contractor 

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States • Department of Psychiatry and Behavioral Sciences, Feinberg School of Medicine, Northwestern University, Chicago United States • Department of Pharmacology, Feinberg School of Medicine, Northwestern University, Chicago United States • Department of Neurobiology, Weinberg School of Arts and Sciences, Northwestern University, Evanston United States

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eLife Assessment

Marshall et al describe the effects of altering metabotropic glutamate receptor 5 activity on activity of D1 receptor expressing spiny projection neurons in dorsolateral striatum focusing on two states - locomotion and rest. The authors examine effects of dSPN-specific constitutive mGlu5 deletion in several motor tests to arrive at this finding. Effects of inhibiting the degradation of the endocannabinoid 2-arachidonoyl glycerol are also examined. Overall, this is a **valuable** study that provides **solid** new information of relevance to movement disorders and possibly psychosis.

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Abstract

Striatal spiny-projection neurons (SPNs) integrate glutamatergic inputs from the motor cortex and thalamus with neuromodulatory signals to regulate motor output. *In vivo* Ca²⁺ imaging has demonstrated that spatially overlapping ensembles of direct and indirect pathway SPNs (dSPNs, iSPNs) are co-active during spontaneous movement. This co-activity is statistically greater among nearby neurons, correlates with behavioral state, and changes in an SPN-type-specific manner under pathological conditions. How this co-activity arises and whether is important for movement are not well understood. Co-activity likely arises from shared excitatory inputs, the strength of which are scaled by well-characterized mechanisms of synaptic plasticity. Here, we show that the Group I metabotropic glutamate receptor 5 (mGluR5), which regulates the strength of corticostriatal synapses, is a key mediator of behaviorally relevant SPN co-activity. Pharmacological modulation of mGluR5 signaling bidirectionally altered movement and co-activity, but not the absolute level of activity in dSPNs. Targeted deletion of mGluR5 in dSPNs recapitulated the effects on spatiotemporal neural dynamics and movement, consistent with a striatum-specific effect of mGluR5

modulation. Targeted deletion of mGluR5 also produced changes in the synaptic properties of dSPNs. Separate from any effects on overall activity, our results show that excitatory synaptic modulation influences motor function by coordinating the spatial co-activation of dSPNs *in vivo*.

Introduction

The dorsal striatum is central to movement initiation and to regulating the kinematics of learned movements^{1,2}. Synaptic plasticity in this input structure of the basal ganglia has been demonstrated to be important for motor learning³. In particular, the strength of corticostriatal synapses between the motor cortex and spiny-projection neurons (SPNs) in the dorsolateral striatum (DLS) are strengthened by learning a motor sequence⁴ or during the acquisition of a motor skill^{5,6}. These glutamatergic, corticostriatal inputs form synapses onto two types of SPNs that are distinguished by their expression of dopamine receptor subtypes and downstream axonal projections. The D1 dopamine receptor-expressing, or direct pathway SPNs (dSPNs) project monosynaptically to the output nuclei of the basal ganglia, the substantia nigra pars reticulata (SNr) and globus pallidus internal segment (GPI). By contrast, D2 dopamine receptor-expressing SPNs (iSPNs) project indirectly to basal ganglia output nuclei via the globus pallidus external segment (GPe) and the subthalamic nucleus (STN)⁷. In classical models of the basal ganglia circuit, activity in dSPNs facilitates movement and activity in iSPNs inhibits movement. Therefore, the relative level of activity between the two populations is predicted to correlate with an animal's motor state (*e.g.*, moving or at rest).

This rate-model framework of striatal function garnered early support from anatomical and pharmacological observations as well as findings from studies investigating the neuropathology of striatum-related neurodegenerative disease⁸. However, the model is inconsistent with more recent observations employing *in vivo* imaging techniques. In these studies, where the activity of genetically defined iSPNs and dSPNs can be monitored concurrently in awake behaving animals, dSPNs and iSPNs are consistently co-active during movement initiation and procession through movement sequences⁹. More recently, it was shown that dSPNs and iSPNs simultaneously activate with spatially clustered dynamics, and that this spatiotemporal organization is important for motor control^{10–12}. Spatiotemporal clustering, or heightened co-activity among proximal SPN pairs, is most pronounced during periods of rest, but also transiently increases during transitions from rest to movement then subsides as SPNs become more active during movement progression^{10–13}. Importantly, dopamine signaling appears to play an important role in sculpting these dynamics. The loss of dopamine diminishes the spatiotemporal coordination of iSPN activity, while heightened dopamine release de-correlates activity in dSPNs and heightens iSPN co-activity^{10–13}.

Although the mechanisms that constrain the spatiotemporal coordination of dSPN and iSPN activity are not precisely known, SPNs have hyperpolarized resting membrane potentials that prevent action potential generation in the absence of glutamatergic input^{14,15}. Given that glutamate is required for SPN activation, and given dopamine's role in modifying the strength of glutamatergic synapses in SPNs, it is likely that synaptic plasticity influences the spatiotemporal dynamics of SPNs by regulating their collective sensitivity to convergent excitatory inputs^{16,17}.

dSPNs and iSPNs receive excitatory input from the cortex and thalamus. Cortico- and thalamostriatal synapses differ in their expression of pre- and post-synaptic neurotransmitter receptors and forms of synaptic plasticity¹⁸. Extensive *in vitro* work has demonstrated various mechanistically distinct types of long-term potentiation (LTP) and long-term depression (LTD) at cortical inputs to dSPNs and iSPNs^{19,20}. While the range of physiological and neuromodulatory mechanisms of corticostriatal plasticity are complex, one well-studied type of

plasticity is endocannabinoid (eCB) dependent LTD, induced *in vitro* by moderate to high frequency stimulation of corticostriatal inputs paired with postsynaptic depolarization¹⁸. Importantly, corticostriatal LTD requires activation of postsynaptic Gq-linked, Group I metabotropic glutamate receptors (mGluRs). mGluR activation causes the mobilization of endocannabinoids (eCBs) from the postsynaptic neuron which act as a retrograde signal by activating presynaptic CB1 receptors to reduce the release probability of glutamatergic corticostriatal synapses²¹. mGluR LTD occurs at corticostriatal synapses in both dSPNs and iSPNs²² but not at thalamostriatal synapses¹⁸. eCB release and retrograde signaling can also produce transient modulation of presynaptic release to cause the short-term depression of corticostriatal synapses^{23,24}, similar to what has been observed at synapses in several other brain regions²⁵. Despite extensive *in vitro* work, little is known about the circumstances under which eCB-dependent synaptic modulation occurs *in vivo* and how it may relate to motor behavior. Moreover, it is unknown how the modulation of corticostriatal synaptic strength regulates the level and/or spatiotemporal dynamics of SPN activity with respect to movement.

In the striatum, mGluR5 has been shown to have a prominent role in mGluR-dependent synaptic plasticity. Therefore, we asked whether mGluR5 signaling contributes to voluntary movement and the dynamics of dSPN activity in the DLS. We found that manipulating mGluR5 signaling had bidirectional effects on movement and the spatiotemporal dynamics of dSPN activity, with minimal effects on their overall levels of activity. Specifically, using *in vivo* Ca²⁺ imaging we found that pharmacological inhibition of mGluR5 signaling produced a reduction in spontaneous movement and heightened the spatially clustered patterns of co-activity in dSPNs, whereas positive modulation of mGluR5 signaling had the opposite effects. Consistent with these findings, conditional knockout of mGluR5 in dSPNs (D1 cKO mice) had minimal effects on the levels of dSPN or iSPN activity but specifically increased the spatiotemporal coordination of dSPN activity and reduced spontaneous locomotion. Collectively, these findings suggest that mGluR-mediated synaptic plasticity mechanisms bidirectionally constrain the spatiotemporal coordination, but not the levels of dSPN activity, into states that correlate with increased or decreased locomotor activity. Our study lays the groundwork for dissociating the separable roles of the activity rates versus patterning of SPN activity in motor control and has implications for correcting disruptions of this patterning in basal ganglia-associated diseases.

Results

dSPNs exhibit spatially clustered activity that varies with movement

There is a strong correlation between spatiotemporal patterns of SPN activity and motor output^{10–12}, but it is not known if synaptic properties modulate this coordinated activity *in vivo*. To assess the activity of dSPNs during spontaneous motor behavior, we implanted gradient refractive index (GRIN) lenses into the dorsolateral striatum of mice expressing Cre-recombinase in dSPNs under the control of the *Drd1a* promoter. A Cre-dependent AAV to express GCaMP6f was introduced either by a prior stereotaxic injection of virus or by coating the lenses with a silk fibroin film containing the AAV (**Figure 1a**)²⁶. The GRIN lens was coupled to a head-mounted miniscope for 1P imaging of somatic Ca²⁺ events²⁷. **Figure 1b** shows an example of a field of view from one mouse and the segmented outlines of neurons after processing of the videos (see **Methods**) and examples of individual dSPN Ca²⁺ dynamics (**Figure 1c**). Consistent with prior observations¹³, Ca²⁺ event rates in dSPNs increased during periods of movement (**Figure 1d, e**). The probability of a Ca²⁺ events in detected cells (defined as fluorescence change of 2.5× the baseline for each cell, see **Methods**) increased during movement (**Figure 1d**) and the Ca²⁺ event rate increased with increasing animal velocity (**Figure 1e**). Previous work has demonstrated that dSPNs and iSPNs are spontaneously co-active in spatially intermingled clusters, a feature that has been proposed to be important for action selection^{10–12,28}. To examine this

spatiotemporal coordination and its modulation by movement state, we measured dSPN co-activity using a Jaccard index of dSPN pairs as a function of the Euclidean distance between the cell pairs in the DLS. This measure captures co-activity between neuron pairs (as opposed to co-inactivity) more specifically than a linear correlation coefficient (**Figure 1f**; **Methods**).¹¹ Across all cell pairs in the field-of-view for each session, a correlation matrix was generated in which, for each frame of the recording, the cell pair was scored a 1 if both cells were active, or 0 otherwise. These scores for each frame were divided by the number of events in both cells to generate a co-activity score for that pair. **Figure 1f** shows schematized raster plots to demonstrate this process. Then, those values were normalized to averages measured from independently shuffled Ca^{2+} traces for each cell in the recording (using 1000 independent shuffles for each cell trace), to control for changes in mean activity level between conditions (**Figure 1f**). If cells were co-active independently of distance, then a plot of co-activity normalized to the shuffled data as a function of distance would be flat (**Figure 1g, top**), while clustered co-activity (i.e., higher among proximal cells) would show a downward sloping curve (**Figure 1g, bottom**). Consistent with prior reports^{10,12}, this analysis revealed that co-activity among neuron pairs is markedly higher among proximal cells and decreases with increasing Euclidean distance between pairs (**Figure 1h, top**). Furthermore, as we had previously demonstrated, co-activity among all pairs was higher during rest and decreased during periods of movement (**Figure 1h, top**).¹³

We next quantified the extent to which activity was higher in proximal cells versus distal neurons in each frame of the recording using a spatial coordination index (SCI), as defined previously¹¹. Here, for each frame of the down-sampled video, we preserved the cell activity traces but shuffled the pairwise distances for each neuron pair. We then performed a pair of Kolmogorov–Smirnov tests (**methods**) to determine if the cumulative distribution of active cell distances in each frame was smaller (i.e., left-shifted or more “clustered”) or greater (right-shifted or less “clustered”) than that of the shuffled data. This analysis allows for visualization of how the spatiotemporal dynamics SPN activity change instantaneously (e.g., during movement onset). Consistent with that reported previously, the degree of “clustered” activity quantified in this way increased significantly during movement onset (within 2.5 seconds of the mouse initiating movement)¹¹. This transient increase contrasts with the increased basal co-activity at all cell-cell distances seen during periods when mice were at rest (defined as centroid velocity below a given threshold).

The spatiotemporal dynamics of dSPN activity is modulated by mGluR5

The dominant form of synaptic plasticity at corticostriatal synapses is mediated by Group1 mGluRs¹⁸, and the primary receptor of this class expressed in SPNs is mGluR5²⁹. Germline deletion of mGluR5 causes significant motor phenotypes, but it is not known how mGluR5 signaling directly affects motor function³⁰. Due to the known importance of mGluR5 in modulating synaptic weight at corticostriatal synapses, we hypothesized that pharmacological modulation of mGluR5 would affect SPN dynamics *in vivo*.

To assess the effects of modulating mGluR5 activity on dSPN dynamics during spontaneous behavior, we injected mice with vehicle or the mGluR5 negative allosteric modulator (NAM) fenobam (26 mg/kg via i.p. injection) immediately prior to a 60-minute open field test. We analyzed dSPN activity from 8 animals expressing GCaMP6f in dSPNs and implanted with GRIN lenses in the DLS (**Figure 2a–c**). We measured activity as mice spontaneously explored the open field arena. On average we recorded from 91 dSPNs per mouse in these experiments.

Fenobam treatment uniformly reduced locomotion (**Figure 2d, e**), an effect that was driven more by a reduction in movement bout duration than the frequency of movement bouts (**Figure 2f**). Interestingly, dSPN event rates were unchanged following fenobam injection, with Ca^{2+}

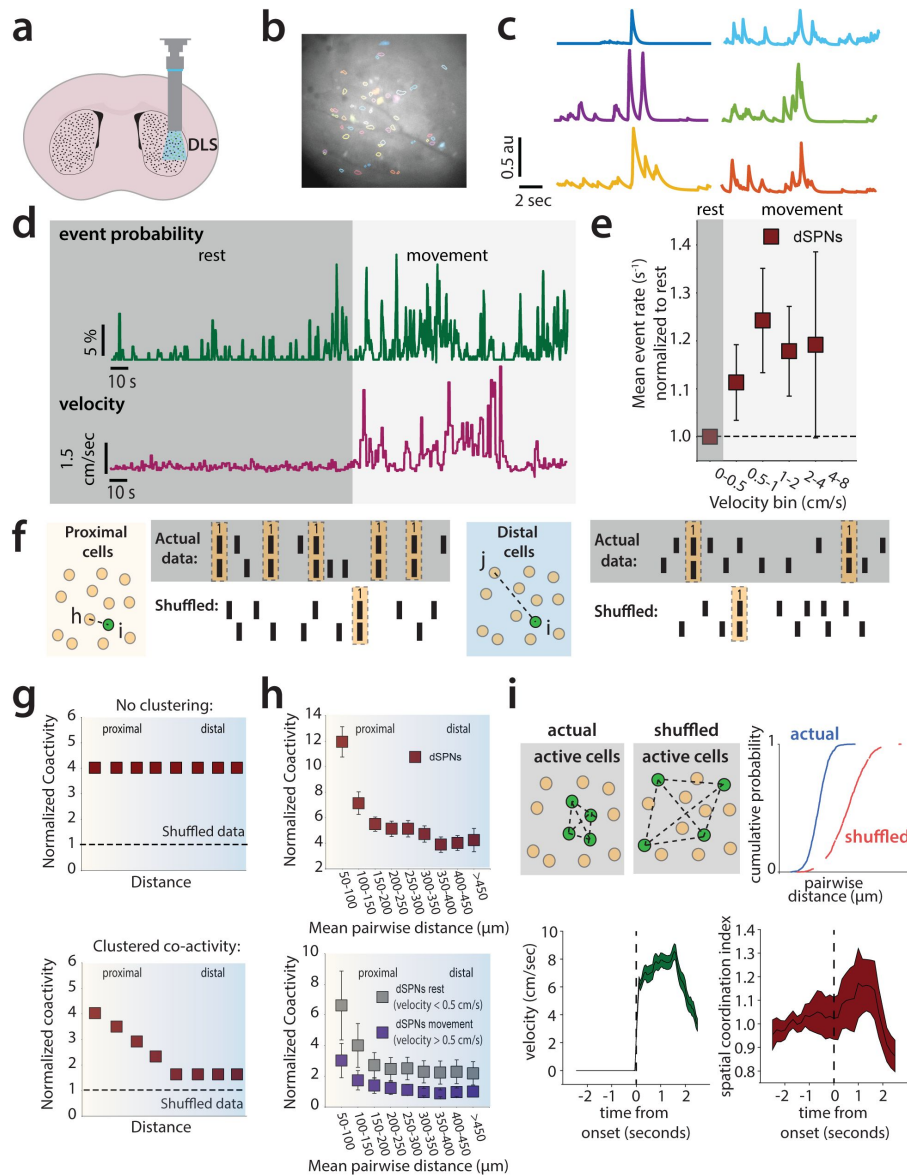


Figure 1.

dSPNs exhibit spatially clustered activity that varies with movement.

(a) Cartoon representation of implanted GRIN lens and miniscope imaging in the DLS. (b) Examples of segmented cell outlines from the CNMF-E algorithm in a single mouse. (c) Example of Ca^{2+} traces from corresponding colored and filled segmented cells shown in (b). (d) The probability of a Ca^{2+} event in detected cells increases during movement in dSPNs. (e) The Ca^{2+} event rate in detected cells increases during movement in dSPNs ($p=0.240$, Wilcoxon Signed Rank Test, events/second during bins with velocity >0.5 cm/sec vs. during bins with velocity <0.5 cm/sec). (f) Cartoon schematic to illustrate analysis of co-activity as a function of distance between cell pairs. Across all cell pairs in the field-of-view for each session, a correlation matrix was generated, in which, for each frame of the recording, the cell pair was scored a 1 if both cells were active, or 0 otherwise. These scores for each frame were averaged across all frames in the recording, to generate an average co-activity score for that pair. Then, those values were normalized to averages measured from independently shuffled Ca^{2+} traces for each cell in the recording (using 1000 independent shuffles for each cell trace), to control for changes in mean activity level between conditions. (g) Hypothetical plot of shuffle-normalized jaccard scores versus distance when coactivity is not “clustered” and hypothetical plot of “clustered” co-activity. Co-activity decreases during periods of movement (Linear Mixed Effects Model/LMM, $p<0.000$, $z=-4.852$, $\beta=-1.671$, $\text{SE}=0.344$, 95% CI: $[-2.346 -0.996]$, $N=5$ mice). (i) The spatial coordination index (SCI) peaks during movement onset ($p<0.000$, $z=12.21$, Wilcoxon Signed Rank Test, comparing the peak value of the normalized SCI during 2.5 seconds following movement onset to 1).

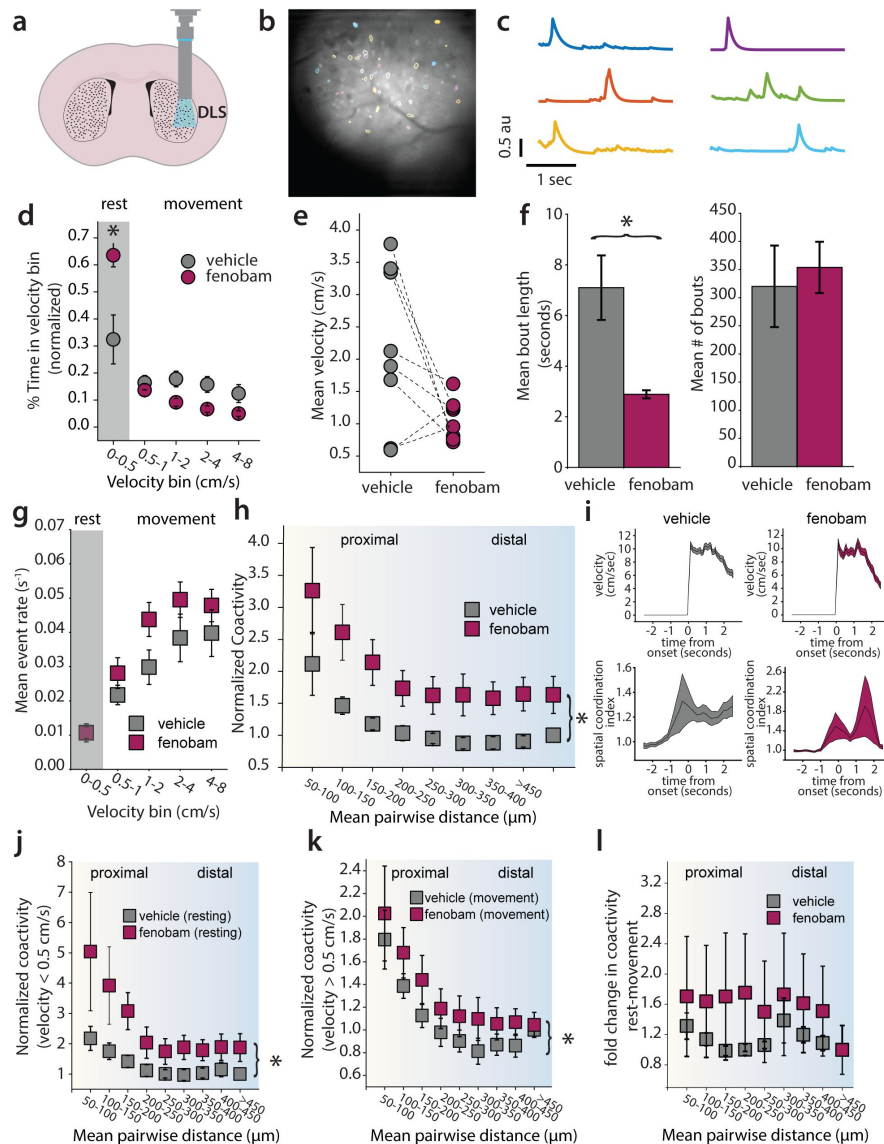


Figure 2.

A negative allosteric modulator of mGluR5 increases coactivity among dSPN pairs.

(a) Cartoon representing implanted GRIN lens and miniscope imaging in the DLS (b) Examples of segmented cell outlines from the CNMF-E algorithm in a single mouse. (c) Example of Ca^{2+} traces from corresponding filled segmented cells shown in (b) (d) Binned velocity of animal. Fenobam increased the amount of time the animal spends at rest (Vehicle mean: 0.324 ± 0.091 , fenobam mean: 0.635 ± 0.043 , $p=0.024$, $U=10$, Mann-Whitney U test) (e) Mean velocity of animals following vehicle or fenobam treatment. The fenobam effect on mean velocity during the open field test is not significant (Mann-Whitney U test, $p=0.103$, $U=48$) (f) Mean bout length of animals during vehicle or fenobam. Fenobam reduces the mean bout length (Vehicle: 7.098 seconds ± 1.272 , Fenobam: 2.886 seconds ± 0.160 , Mann-Whitney U test, $p<0.001$, $U=0$), but does not affect the number of bouts (Mann-Whitney U test, $p=0.793$, $U=35$). (g) The average event rate of dSPNs at each movement velocity of the mouse. This relationship is not affected by fenobam administration (Linear Mixed Effects Model/LMM, $p=0.754$, $z=-0.313$, $\beta=-0.002$, $SE = 0.006$, 95% CI: $[-0.014, 0.010]$, $N=8$ mice). (h) Fenobam increases the pairwise coactivity of dSPNs when grouped into $50 \mu m$ bins (LMM, $p<0.000$, $z=-6.050$, $\beta=-0.828$, $SE = 0.137$, 95% CI: $[-1.096, -0.560]$, $N=8$ mice) (i) Following both vehicle and drug treatment spatial coordination peaks during movement onset (Veh: $p<0.000$, $z=12.21$, Drug: $p<0.000$, $z=10.1804$) (j) Pairwise coactivity during rest with vehicle and fenobam treatment (velocity $<0.5 \text{ cm} \cdot \text{sec}^{-1}$) (LMM, $p<0.000$, $z=-4.448$, $\beta=-1.292$, $SE = 0.290$, 95% CI: $[-1.861, -0.723]$, $N=8$ mice) (k) Pairwise coactivity during movement (velocity $>0.5 \text{ cm} \cdot \text{sec}^{-1}$) with vehicle and fenobam (LMM, $p=0.008$, $z=-2.673$, $\beta=-0.220$, $SE = 0.082$, 95% CI: $[-0.381, -0.059]$, $N=8$ mice) (l) Fenobam does not affect the relative decrease in coactivity seen during movement

event rates in dSPNs increasing with spontaneous locomotor speed across the range of running speeds in both fenobam and vehicle injected animals (**Figure 2g**). Fenobam also had no effect on the amplitude or duration of Ca^{2+} events (Figure 2 supplement).

However, fenobam treatment significantly increased the pairwise co-activity of dSPNs across the whole range of cell-cell distances (**Figure 2h**). Notably, while co-activity at all distances was enhanced by fenobam treatment, we did not resolve any differences in spatial coordination, measured using the SCI, between vehicle and drug treatment as animal transitioned from rest to movement. In both cases, the SCI peaked around the time of movement onset (**Figure 2i**). This suggests that the most significant effect of mGluR5 on dSPN co-activity is to increase its overall extent, rather than the relative extent to which it becomes more or less “clustered” during behavioral state transitions. As fenobam treatment reduced movement, and co-activity is modulated by movement, we separated our analysis into periods of movement and rest to compare equivalent behavioral states during vehicle and drug treatment. While fenobam treatment increased dSPN co-activity during both rest and movement, its effects were most pronounced during periods of rest (**Figure 2j, k**). However, in analyzing the relative increase in coactivity between rest and movement, there was no significant difference between fenobam and vehicle treated animals (**Figure 2l**). Taken together these results suggest that fenobam may suppress movement by transitioning spatiotemporal dSPN ensemble dynamics towards a rest-associated state of high overall co-activity, without affecting the absolute rates of their activity. In this rest-associated state, overall neuronal activity is reduced but more correlated, resulting in a higher degree of co-activity.

An mGluR5 positive allosteric modulator (PAM) increases movement and decreases SPN co-activity

Next, we sought to determine how promoting mGluR5 signaling affects motor output and SPN dynamics. We selected JNJ-46778212/VU0409551 (JNJ) an mGluR5 PAM with demonstrated *in vivo* efficacy and known pharmacokinetics in mice³¹. We administered JNJ (100 mg/kg) or vehicle via i.p. injection 30 minutes prior to measuring dSPN dynamics for one hour during locomotion in an open field arena. In contrast to fenobam, JNJ treatment decreased the amount of time the animal spent at rest (**Figure 3a**). However, there was no statistically significant effect on mean velocity (**Figure 3b**), number of bouts (**Figure 3c**), or bout length (**Figure 3d**). Like fenobam, JNJ did not affect the levels of dSPN activity when measured as a function of the animal's running speed (**Figure 3e**). Additionally, there was no effect on the amplitude or duration of Ca^{2+} events (Figure 3 supplement). However, JNJ significantly decreased the co-activity among dSPN cell pairs measured across a range of Euclidean distances (**Figure 3f**). Again, as with fenobam, animals from both the vehicle and drug treatment groups showed significant increases in spatial coordination surrounding movement onset (**Figure 3g**). The effect on overall co-activity was most pronounced, and statistically significant, during periods of rest (**Figure 3i, j**). Therefore, JNJ treatment had effects that were qualitatively inverse to the effects of fenobam on locomotion and dSPN co-activity during periods of rest. These findings are consistent with the idea that JNJ promoted movement by inducing movement-associated dSPN ensemble dynamics by de-correlating their activity.

mGluR5 activation in the striatum results in the mobilization of eCBs, specifically 2-AG, in dSPNs^{21,23,25}. Therefore, we asked whether enhancing eCB levels by preventing 2-AG degradation using the monoacylglycerol lipase (MAGL) inhibitor JZL-184 affected the spatiotemporal dynamics of dSPN activity. We administered to mice JZL-184 (10 mg/kg i.p.) before recording dSPN Ca^{2+} activity for 60 minutes in an open field arena. Interestingly, in contrast to the mGluR5 PAM, JZL-184 significantly reduced rate of Ca^{2+} events (**Figure 3k**) but did not affect the co-activity of dSPNs (**Figure 3l**). These results suggest that broadly enhancing eCB levels does not recapitulate the effects of potentiating mGluR5 signaling specifically.

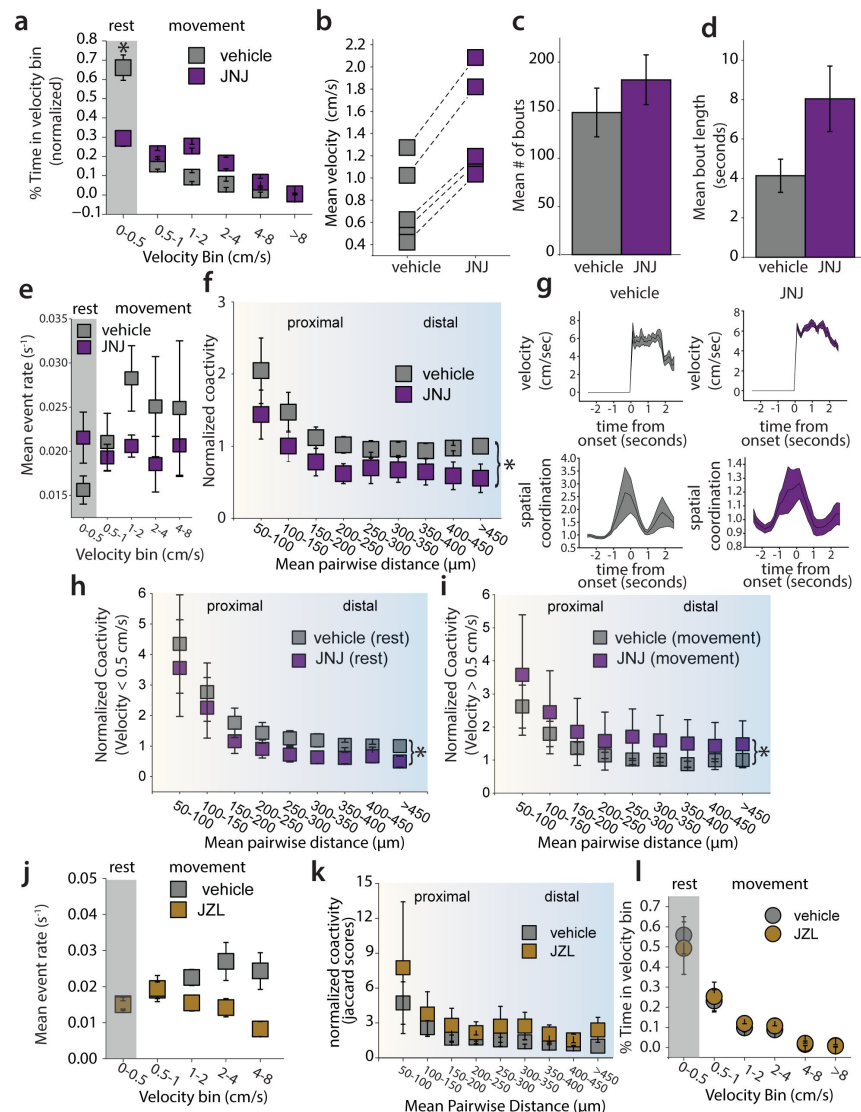


Figure 3.

A positive allosteric modulator of mGluR5, JNJ-46778212, decreases clustered coactivity of dSPNs.

(a) Time animal spent in each velocity bin and (b) Mean velocity of the animals during vehicle and after JNJ administration. JNJ decreases the amount of time the animal spends at rest (Vehicle mean: 0.661 ± 0.066 , JNJ mean: 0.294 ± 0.041 , $p=0.012$, $U=25$, Mann-Whitney U test), however the change in mean velocity (b) is not significant ($p=0.060$, $U=3$, Mann-Whitney U test), nor is the mean bout length during vehicle and JNJ ($p=0.060$, $U=3$, Mann-Whitney U test) (d) Comparison of mean number of bouts of each animal during vehicle and JNJ ($p=0.403$, $U=8$, Mann-Whitney U test). (e) Mean event rate of Ca²⁺ transients in dSPNs during movement. JNJ administration does not significantly affect the event rate in dSPNs (LMM, $p=0.058$, $z=1.893$, $\beta=0.010$, SE = 0.005, 95% CI: [-0.000 0.020], N=4 mice) (f) Coactivity of pairs of dSPNs (50 μm bins) during total time in open field. Systemic JNJ administration reduces pairwise coactivity (LMM, $p=0.003$, $z=2.964$, $\beta=-0.450$, SE = 0.152, 95% CI: [-0.153 0.748], N=4 mice) (g) Spatial coordination increases at movement onset in both vehicle and drug groups (VEH: $p<0.000$, $z=6.249$, JNJ: $p<0.000$, $z=9.111$, Wilcoxon Signed Rank test) (i) Normalized co-activity of dSPNs during rest. JNJ produces a small but significant reduction in coactivity measured during rest (velocity < 0.5 cm*sec⁻¹) (LMM, $p=0.030$, $z=2.165$, $\beta=0.618$, SE = 0.285, 95% CI: [0.059 1.177], N=4 mice) (j) Normalized co-activity of dSPNs during movement (velocity > 0.5 cm*sec⁻¹). JNJ had a significant effect on dSPN clustered activity (LMM, $p=0.027$, $z=-2.212$, $\beta=-1.694$, SE = 0.766, 95% CI: [-3.195 -0.193], N=4 mice). (k) Event rate in dSPNs during vehicle and after JZL administration. JZL-184 administration reduces the event rate (LMM, $p=0.001$, $z=3.195$, $\beta=0.007$, SE = 0.002, 95% CI: [0.003 0.012], N=5 mice) (l) Normalized co-activity of dSPNs against Euclidean distance. JZL had no effect on coactivity (LMM, $p=0.065$, $z=-1.847$, $\beta=-1.204$, SE = 0.652, 95% CI: [-2.481 0.074], N=5 mice) (m) Time animals spent in velocity bins. JZL did not affect time spent at rest ($p=0.531$, $U=16$, Mann-Whitney U test).

Collectively these results demonstrate that potentiating mGluR5 signaling shifts dSPN dynamics to a less correlated state, while inhibiting mGluR5 signaling enhances the correlated state of dSPNs. It is notable that spontaneous dSPN activity is less correlated during periods of movement than rest, and de-correlated dSPN activity has been shown to occur in hyperkinetic states (*e.g.*, dyskinetic and hyperdopaminergic states ^{11,13}). These observations suggest that inhibition of mGluR5 signaling suppresses movement by altering the spatiotemporal dynamics of dSPNs through synaptic mechanisms that alter their collective recruitment via excitatory afferents.

mGluR5 reduces corticostriatal glutamatergic transmission at dSPN synapses in the dorsolateral striatum

Our experiments demonstrated changes in striatal dSPN network activity that correlate with changes in motor behavior mediated by pharmacologically modulating mGluR5. To determine whether modulating mGluR5, specifically in striatal dSPNs, mediates the observed effects on locomotor and striatal activity, we used a genetic approach to selectively manipulate mGluR5 in D1 receptor-expressing neurons. To do this, we crossed *Grm5*^{fl/fl} mice ³² to D1-Cre mice ³³ to create conditional knockout mice (D1 cKO) mice. Further crossing with a Cre reporter allele (Ai9) ³⁴ enabled us to perform patch-clamp recordings from visually identified dSPNs in striatal slices to record corticostriatal EPSCs (**Figure 4a**). As we have previously demonstrated ²⁴, a ten-minute application of the group 1 mGluR agonist DHPG (100 μ M) caused a depression of EPSCs in WT recordings (**Figure 4b**). This effect was significantly diminished in recordings from cKO mice, demonstrating a disruption of mGluR5 signaling (**Figure 4c**).

A number of behavioral phenotypes have been described in mice with constitutive, germline *Grm5* deletion ³⁰. Given our previous observations and the fact that SPN activity in the dorsal striatum is important for motor control, we asked whether selective ablation of *Grm5* in dSPNs affected locomotion and motor learning. Tracking distance traveled in the open field, we found that the D1 cKO mice had reduced spontaneous locomotion (**Figure 4d**) and displayed a reduction in total distance traveled (**Figure 4e**). We also found the D1 cKO mice had diminished digging behavior in a novel environment (**Figure 4f**). Taken together these results demonstrate that selectively disrupting mGluR5 signaling in dSPNs reduces spontaneous movement in novel environments.

To measure locomotion in the home cage setting, we introduced running wheels and monitored activity over 7 days. During the first 3 days, D1 cKO mice demonstrated significantly reduced voluntary running (**Figure 4h, i**). However, both control and D1 cKO mice increased the time spent running over the course of 7 days such that the difference between the groups was not significant by day 7 (**Figure 4i**).

Finally, on an accelerating rotarod D1 cKO mice showed a consistently reduced latency to fall over the course of 12 training trials across three days (**Figure 4g**). Despite this deficit, D1 cKO mice did improve over the course of training, indicating that they were capable of learning the task. Thus, the D1 cKO mice exhibited a primary deficit in spontaneous motor activity with only minor deficits in the learned motor tasks of running on a wheel or forced locomotion on the rotarod.

Because of the clear motor phenotype in D1 cKO mice, we determined whether there were any changes in the intrinsic or synaptic properties of dSPNs in these mice. Whole-cell current-clamp recordings from visually identified dSPNs expressing tdTomato found no difference in the voltage response to increasing current injection suggesting no alteration in the intrinsic excitability of dSPNs in D1 cKO mice (**Figure 4j, k**). In recordings of mEPSCs, that reflect all excitatory synapses (corticostriatal and thalamostriatal), we found that the amplitudes of mEPSCs were unchanged, but their frequency was significantly increased in dSPNs from D1 cKO mice (**Figure 4l-m**), indicating increased release probability when mGluR5 expression is disrupted.

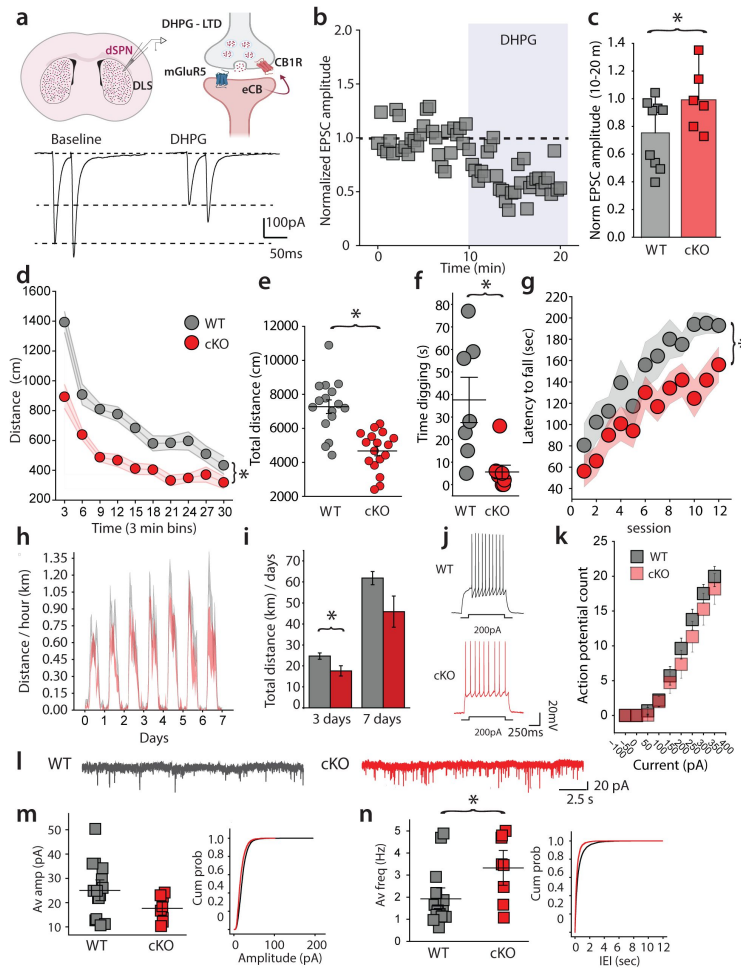


Figure 4.

mGluR5 cKO in *Drd1a*-expressing neurons reduces spontaneous motor behaviors and affects synaptic properties of dSPNs.

(a) (Top) Cartoon representation of the role of mGluR5 in mobilizing eCBs at corticostriatal synapses. (Bottom) Representative EPSCs recorded prior to and during DHPG in WT mice (b) Example time course of corticostriatal EPSCs recorded in dSPNs during application of 100 μ M DHPG in slice from WT mice. (c) Grouped data of DHPG depression. D1 cKO mice lack DHPG induced synaptic depression (cKO normalized EPSC amplitude: 0.754 ± 0.082 , Wilcoxon Signed Rank Test, $p=0.018$, $W=2$; WT normalized amplitude: 0.992 ± 0.992 , Wilcoxon Signed Rank Test $p=0.834$, $W=9$). (d) Time course of activity in open field and (e) Total distance traveled by WT and cKO mice. D1 cKO mice have reduced activity in the open field, measured as distance traveled over time and total distance traveled during a 30-minute open-field session (Mann-Whitney U Test, $U=245$, $p<0.001$, $N = 16$ WT, 17 cKO). (f) Digging time in WT and cKO mice. D1 cKO mice spend less time digging when placed into a novel home cage (Mann-Whitney U Test, $U=51.5$, $p=0.008$, $N = 7$ WT, 8 cKO). (g) Latency to fall in the accelerating rotarod test. dSPN cKO mice have consistently reduced latency during the accelerating rotarod test over 3 days, with 4 sessions per day ($p=0.024$, $F=5.709$, Two-way repeated measures ANOVA, $N = 12$ WT, 17 cKO). (h) Running distance per hour on wireless wheels placed in home cage over 7 days. (i) Total distance run per day during first 3 days and over 7 days. D1 cKO mice ran significantly less during the first 3 days of testing (Mann-Whitney U Test, $U=173$, $p=0.049$) (j) Example traces of voltage response to 200pA current injection in WT and cKO mice. (k) There is not significant difference in the input output curve for current injection and AP firing between genotypes (LMM, $p=0.056$, $z=1.913$, $\beta=1.197$, $SE = 0.626$, 95% CI: $[-0.029, 2.424]$, $N=26$ cells, WT, $N=22$ cells, cKO). (l) Example mEPSC recordings in dSPN from WT and D1 cKO mice (m) Average amplitude of mEPSCs and cumulative distribution of amplitudes in WT and D1 cKO mice. There is no difference in amplitudes in recordings from the two genotypes (WT AMP: 21.04 ± 1.71 pA, $N=15$, cKO AMP: 16.66 ± 1.12 , $N=8$, Mann-Whitney U Test, $U=86.5$, $p=0.093$) (n) Average mEPSC frequency and cumulative interevent intervals in WT and D1 cKO animals. D1 cKO mice have increased mEPSC frequency (WT Hz: 1.920 ± 0.3319 pA, $N=15$, cKO Hz: 3.325 ± 0.525 , $N=8$, Mann-Whitney U Test, $U=27.5$, $p=0.039$)

Spatiotemporal activity of dSPNs is disrupted in D1 cKO mice

To measure the *in vivo* dynamics of dSPNs in D1 cKO mice, we implanted GRIN lenses into the DLS of control and D1 cKO mice and analyzed dSPN Ca^{2+} activity during 20 minutes of open field exploration (**Figure 5a–e**). We performed control imaging experiments in mice expressing Cre in dSPNs but with the wild type *GRM5* allele (D1 cre; *GRM5* +/+). Consistent with our previous results, the event rate of dSPN Ca^{2+} transients increased with locomotor speed in both experimental and control mice (**Figure 5d**). Selective mGluR5 deletion in *Drd1a*-expressing neurons did not alter the rate of dSPN Ca^{2+} events compared to control mice (**Figure 5d**). cKO mice and control animals also showed similar distributions of event amplitudes and durations (Figure 5 supplement). Instead, the normalized coactivity of dSPNs (analyzed across 50- μm bins of pairwise cell distances) was significantly increased in D1R cKO mice, similar to the effect of the mGluR5 NAM fenobam (**Figure 5e**). Additionally, D1R cKO mice and control mice showed a transient peak in coactivity surrounding motion onset (**Figure 5f**). When measured over all frames, the coactivity in both control and cKO mice increased during continuous period of rest compared to periods of continuous movement, however there was no difference in this effect between groups (**Figure 5g**). Therefore, targeted genetic deletion of mGluR5 in dSPNs had qualitatively similar effects as systemic administration of the mGluR5 NAM on co-activity of dSPNs.

Finally, because both dSPNs and iSPNs exhibit spatially clustered activity *in vivo*, and because they are synaptically connected through recurrent collaterals³⁵, it is possible that disruption of mGluR5 signaling in dSPNs may also affect the levels and spatiotemporal dynamics of activity in intermingled iSPNs. To address this question, we used a viral “Cre-off” strategy to express GCaMP6 in Cre-negative, iSPNs of *Drd1a*-Cre; *Grm5*^{fl/fl} mice³⁶. We found that the deletion of mGluR5 in dSPNs had no effects on either the levels of activity in iSPNs (**Figure 5h**) or their spatiotemporal clustering (**Figure 5i**). Therefore, the alteration in SPN dynamics in D1 cKO mice appear to be cell-autonomous, as specific targeting of mGluR5 in dSPNs did not affect iSPN activity in D1 cKO mice.

Together, these results demonstrate that the manipulation of mGluR5 signaling, through pharmacological or targeted genetic means, leads to changes in the spatiotemporal patterns of activity in dSPNs with minimal effects on their overall levels of activity. Notably, these spatiotemporal dynamics varied with locomotor speed under normal conditions, and manipulations of mGluR5 signaling induced changes in these dynamics that correlated with locomotor state. These findings suggest that these two facets of activity—levels and spatiotemporal patterning of activity—are separable and provide important insights into the role of spatiotemporally coordinated activity in the generation of movement.

Discussion

The rate-model of basal ganglia function posits that activity in dSPNs and iSPNs have distinct and opposing effects on movement. This framework is primarily based on anatomical evidence from the distinct projection targets of dSPNs and iSPNs and electrophysiological recordings in their downstream projection targets under conditions modeling parkinsonism⁸. These target nuclei inhibit motor control nuclei in the hindbrain mesencephalic locomotor region and thalamic regions that project to the motor cortex^{37,38}. In this context, dSPN activation is predicted to increase activity in these motor control regions to facilitate movement, while iSPN activation is predicted to decrease activity in these same regions to inhibit movement.

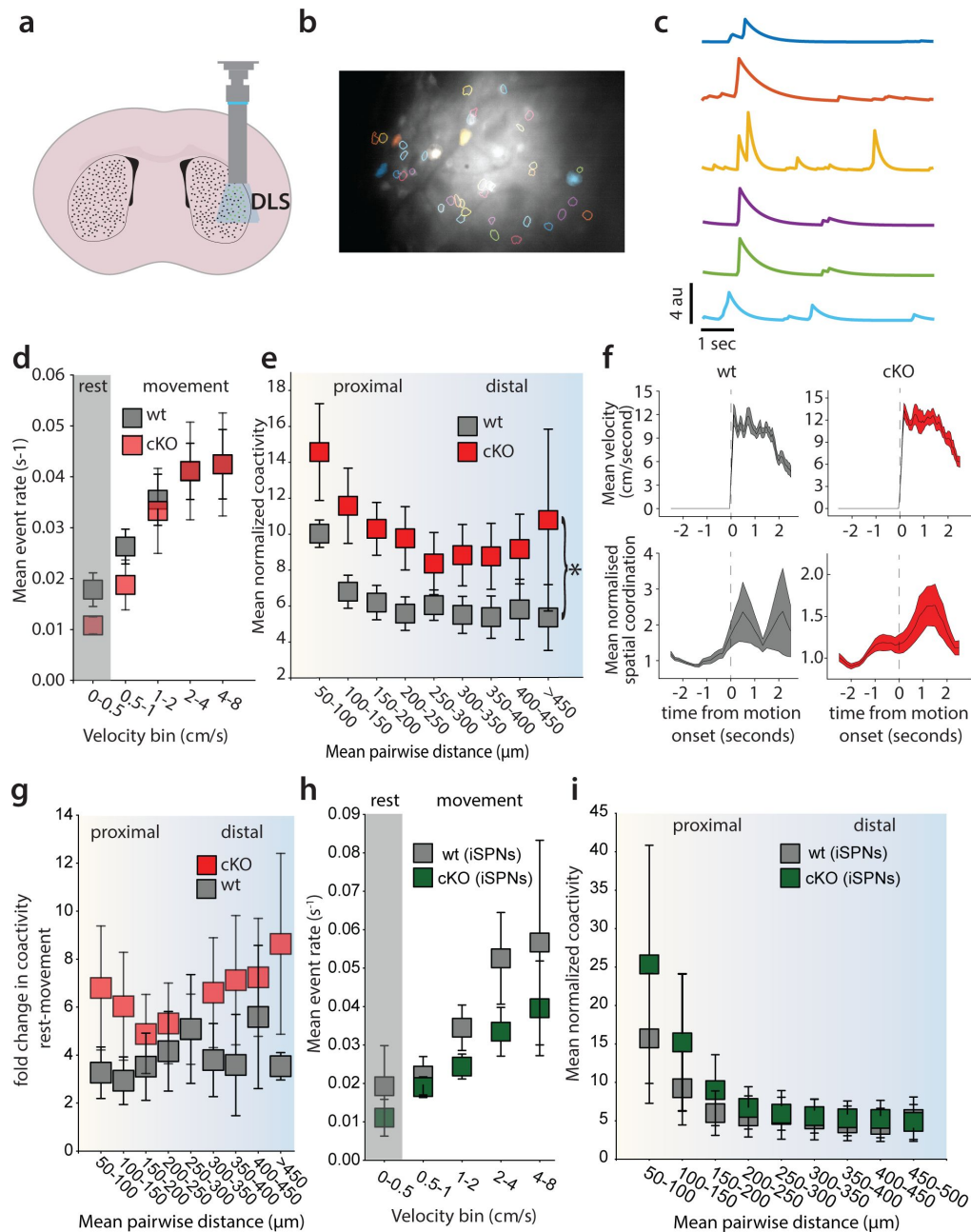


Figure 5.

D1 cKO mice show cell autonomous increase in clustered coactivity in dSPNs.

(a) Cartoon representation of miniscope imaging in the DLS (b) Example segmented image from cKO mice (c) Example Ca^{2+} activity traces from identified dSPNs in cKO mice, filled cells in (b) (d) Mean event rate during movement in WT and cKO animals. The event rate in dSPN cKO mice is unchanged from WT (LMM, $p=0.651$, $z=0.452$, $\beta=0.003$, $\text{SE}=0.007$, 95% CI: $[-0.011, 0.017]$, $N=9$ wt, 6 cKO) (e) Coactivity of pairs of dSPNs (50 μm bins) in WT and cKO animals. D1 cKO mice have increased coactivity across all pairwise distances (MLM, $p=0.021$, $z=2.305$, $\beta=3.989$, $\text{SE}=1.731$, 95% CI: $[0.597, 7.382]$, $N=9$ wt, 6 cKO) (f) Spatial coordination peaks at movement onset in both wt and cKO mice (WT: $p<0.000$, $z=7.301$, KO: $p<0.000$, $z=6.700$, Wilcoxon Signed Rank test) (g) The relative increase in coactivity during rest is not statistically different between cKO and wt mice (LMM, $p=0.552$, $z=0.595$, $\beta=1.391$, $\text{SE}=2.339$, 95% CI: $[-3.193, 5.976]$) (h) iSPN event rate in D1 cKO and WT animals is unchanged (LMM, $p=0.310$, $z=1.016$, $\beta=0.017$, $\text{SE}=0.016$, 95% CI: $[-0.015, 0.048]$, $N=3$ wt, 3 cKO) (i) Clustered coactivity of iSPNs in WT and D1 mGluR5 cKO animals is unaffected (LMM, $p=0.691$, $z=0.398$, $\beta=2.368$, $\text{SE}=5.953$, 95% CI: $[-9.300, 14.035]$, $N=3$ wt, 3 cKO).

However, this classical model is not supported by recent *in vivo* imaging studies using fiber photometry or endoscopy to record Ca^{2+} activity in genetically specified dSPNs or iSPNs. In contrast to the predictions of the rate-model, dSPNs and iSPNs are indistinguishably responsive to increases and decreases in movement ⁹. Intriguingly, *in vivo* endoscopy has shown that overlapping clusters of dSPNs and iSPNs co-activate during movement, with small but statistical correlations between the activation of different clusters during different movement types ¹⁰. These findings suggest an alternative model where, in contrast to the relative level of activity in each SPN type, the degree of co-activity among SPNs and ensemble identity may be important for specifying proper motor output.

Despite evidence that SPNs may influence movement via the coordination of their activity, the biological mechanisms responsible for these coordinated patterns are not known. Seminal *in vitro* electrophysiology studies demonstrated that SPNs fire action potentials coincident with large excitatory inputs that shift their resting membrane potential to transiently stable, depolarized “up” states ^{14,15}. These excitatory inputs arise from the cortex and thalamus. Our results suggest that the modulation of the synaptic weight, particularly at corticostriatal synapses where several forms of plasticity require mGluR5 receptors, is a crucial mechanism constraining the patterns of dSPN co-activity in states associated with movement or rest.

Spontaneous motor output correlates with the level of coactivity in dSPNs

In the current work, we used pharmacological and genetic manipulations to examine how activity in dSPNs varies with movement while targeting the receptor controlling the dominant form of synaptic plasticity in the DLS. Systemic administration of the mGluR5 negative allosteric modulator, fenobam, led to significant increases in co-activity among dSPNs and reduced locomotor activity. By contrast, systemic administration of the mGluR5 positive allosteric modulator JNJ decreased dSPN co-activity and increased locomotion. Notably, neither pharmacological manipulation of mGluR5 led to significant changes in the rate of dSPN Ca^{2+} events. Our results suggest a model whereby either potentiating or inhibiting mGluR5 signaling alters the recruitment of dSPNs such that the overall activity level is preserved but that the patterns of their co-activity change in a manner that corresponds to motor output. This is further confirmed by our data from animals with targeted deletion of mGluR5 in dSPNs, which similarly showed no change in event rate but an increase in co-activity.

Our results are consistent with prior analysis demonstrating that the co-activity among pairs of dSPNs is higher at rest than during movement ¹³. Here we found that pharmacological manipulations that alter these clustered co-activity states correspondingly alter the levels of spontaneous motor output. Specifically, shifting the dSPN network to a level of higher co-activity, as with fenobam treatment or mGluR5 deletion in dSPNs, leads to a reduction in the amount of time animals spent moving. By contrast, potentiating mGluR5 receptors with systemic JNJ administration led to an increase in time spent moving and a corresponding decrease in the dSPN co-activity.

Our *ex vivo* electrophysiology data from cKO animals further supports these conclusions, as disrupting endogenous mGluR5 signaling in dSPNs does not result in changes in the excitability of neurons. Instead, genetic deletion of mGluR5 altered the synaptic properties of dSPNs, likely an adaptive response consistent with the established role of mGluR5 in regulating corticostriatal synaptic weights. These findings suggest that these selective changes in synaptic plasticity may underlie the altered spatiotemporal patterning of dSPN activity without changes in their overall levels of activity observed *in vivo*.

It is important to note that, in our pharmacological experiments, we observed that inhibition of mGluR5 with systemically administered fenobam increased the co-activity among active neuron pairs across all distance bins we measured. That is, co-activity was increased in both proximal and distal neuron pairs and this increased co-activity correlated with decreased ambulatory behavior. Administration of an mGluR5 PAM decreased co-activity when measured across all binned distances of neuron pairs and this occurred in coincidence with increased ambulatory behavior. It has previously been demonstrated¹¹, and we replicated here, that surrounding movement onsets there is an increase in the extent to which proximal neuron pairs are statistically more likely to be co-active than distal neuron pairs. Using a previously established metric for quantifying this increase, the spatial correlation index, we did not resolve any significant changes in the onset related changes in this metric following either genetic or pharmacological alterations of mGluR5 activity. Our central finding, observed with both pharmacological and genetic manipulations, is that manipulating mGluR5 signaling alters overall dSPN co-activity, irrespective of cell proximity, in a manner that correlates with changes in motor output. We hypothesize that these changes are likely due to alterations in synaptic weight at dSPN synapses that occur following mGluR5 manipulation.

One way in which mGluR5 may modify synaptic weight is through long-term plasticity. *In vitro* studies have demonstrated that mGluR5 activation occurs following pairing of postsynaptic depolarization with moderate to high frequency excitatory synaptic input²¹. mGluR5 activation can result in the long-term depression (LTD) of synaptic transmission or a more transient depression in release probability^{23,24,39}. If rest associated ensembles of dSPNs are substantially different in composition from movement and/or movement-onset ensembles, then our data suggests that synapses onto dSPNs may have a lower synaptic weight during spontaneous behavior than movement or movement onset ensembles, potentially due to eCB release. Inhibiting mGluR5, either through pharmacology or genetics, may increase this synaptic weight, thereby promoting activation of rest ensembles and decreasing motor output. This input that primarily governs the activation of movement-related ensembles could either be a separate population of cortical inputs with higher synaptic weight or thalamostriatal inputs more resilient to mGluR5 manipulations¹⁸.

Our results are an important step in understanding how the regulation of synaptic weights by mGluR5 may act on the coordinated activity of dSPNs *in vivo* to regulate behavior.

Global enhancement of eCB levels does not alter clustered co-activity

The lack of change in spatiotemporal coordination following systemic administration of the 2-AG degradation inhibitor JZL-184 suggests that global enhancement of eCB levels does not produce a similar outcome as modulating mGluR5 signaling. However, in contrast to selective mGluR5 modulation we did observe a reduction in overall dSPN event rates with JZL. One possibility is that a degree of synaptic specificity is required to affect clustered activation of dSPNs. Thus, JZL treatment may have exerted such broad effects on the circuit that it resulted in a different outcome for neural clustering and motor output than that observed following treatment with the allosteric modulators of mGluR5 receptors that modify the weight of specific active synapses.

These results support a model whereby the level of co-activity among specific groups of dSPNs is potentially important for controlling motor output. Therefore, globally altering the synaptic weights of all neurons may overshadow the results of synaptic alterations in specific subsets of SPNs. This idea is consistent with the notion that individual dSPNs may be precisely tuned to have specific roles in regulating motor output, such that altering the activity of one dSPN subgroup would not have the same effect as altering the activity of a different sub-group. This interpretation is also at odds with the classical rate-based models of SPN activity.

Role of non-dSPN mGluR5 receptors in regulating movement

There are notable differences in the design of our experiments using conditional deletion of mGluR5 from dSPNs and systemic administration of a negative allosteric modulator of mGluR5. Following fenobam administration, we observed a significant increase in co-activity when animals were at rest and a smaller increase during movement, however the relative increase in coactivity between rest and movement did not differ significantly from controls in either cKO animals or following fenobam treatment. Therefore, in both cases, inhibition of mGluR5 may promote the activation of rest ensembles and thereby inhibit motor output.

Within the striatum alone, mGluR5 is expressed in iSPNs and interneurons and modulation of these cell types by systemically administered compounds could produce changes in behavior and striatal circuit function. dSPNs receive GABAergic input from both iSPNs as well as striatal interneurons. Of particular note, parvalbumin-positive fast-spiking interneurons also receive excitatory input from the cortex and provide feedforward inhibition onto both subtypes of SPNs, and genetic disruption of mGluR5 in parvalbumin interneurons increases striatum-dependent repetitive behaviors⁴⁰. The modulation of synaptic strength at these connections by fenobam and JNJ could also partly account for the effects of these compounds on dSPN activity and behavior when systemically administered, which would not occur in cKO animals.

Collateral inhibition between SPNs is another potential source for modulation of co-activity. *In vitro* electrophysiology has shown that dSPNs are more likely to form synapses onto other dSPNs, while iSPNs form synapses onto neurons of both pathways with equal probability³⁵. However, the strength of collateral synapses formed by dSPNs appears to be smaller than that of synapses formed by iSPNs. Our analysis of iSPN activity in cKO mice suggests that iSPN circuit properties are largely normal when mGluR5 is genetically disrupted in dSPNs, perhaps reflecting the less substantial modulation of iSPN circuitry by dSPNs reported in the striatum, at least during spontaneous behavior. Whether this relationship holds across additional behavioral states remains to be addressed. Additionally, whether disruption of iSPN networks affects dSPN networks, perhaps due to stronger unidirectional iSPN-dSPN connectivity, is a key question for further research.

Translation implications of dSPN spatiotemporal dynamics

The relationship of mGluR5 pharmacology to dSPN spatial coordination and motor output is clinically relevant to the treatment of motor disorders. Prior studies examining the spatial coordination of SPNs in parkinsonian mice demonstrated that, following dopamine depletion to model a parkinsonian state, dSPN activity was reduced but maintained spatial coordination¹¹. However, when levodopa induced dyskinesia (LID) was modelled in these animals, dSPNs lost spatial coordination. In the current work, we show that pharmacological inhibition of mGluR5 increases spatial coordination. It is tempting to speculate, therefore, that in the case of LID, the therapeutic efficacy of mGluR5 NAMs may involve regulation of the spatially coordinated co-activity of dSPNs. Additionally, further examination of how spatially coordinated SPN activity is disrupted in motor diseases involving basal ganglia pathology, may be a promising avenue for future therapeutic development.

In summary our data demonstrate that the modulation of signaling that affects synaptic properties of excitatory synapses onto dSPNs is a key mediator of the normal patterns of SPN co-activity that typify the striatum's neural ensemble dynamics in relation to spontaneous movement. These results provide further support for models where the local synchronization of SPNs (and the basal ganglia circuit in general) is a stronger determinant of motor output than their overall levels of activity. Our results also provide a framework for future lines of inquiry to address how striatal synaptic properties shape circuit function and striatum-dependent behavior.

Methods

Mice

Animals were group housed with 14-hr:10-hr Light/Dark cycle and food and water were provided ad libitum. Mice of both sexes, between 4-12 months old, were used and all experiments were performed in accordance with procedures approved by the Northwestern University IACUC. To express GCaMP6f in dSPNs we used mice expressing Cre recombinase under the control of the *drd1a* promoter (D1cre, *Drd1a*-Cre Ey262) ⁴¹. A Cre-dependent GCaMP6f under the control of the synapsin promoter (pAAV.Syn.Flex.GCaMP6f.WPRE.SV40, 100833-AAV9 from Addgene) was used for delivery of the construct. For in vitro patch clamp electrophysiology experiments we further crossed the D1Cre line to mice expressing a Cre-dependent tdTomato reporter inserted in the *Gt26Sor* (*Rosa26*) locus (Ai9), in order to identify dSPNs in the DLS ⁴².

For genetic deletion of mGluR5 we used floxed mGluR5 mice (mGluR5^{loxP/loxP}) ³² crossed to either D1cre mice (D1cre; mGluR5^{f/f}) for in vivo imaging experiments, or D1 cre mice crossed with Ai9 mice (D1cre; mGluR5^{f/f}; Ai9^{+/+}) for patch clamp experiments. All mice were maintained on the C57BL6/J background strain.

Behavioral analysis

Motor learning on the accelerating rotarod

Mice of both sexes were placed on the rotarod and the initial revolution speed set to 4 rpm for 120 s. The rotarod then accelerated at a rate of 0.12 rpm·s⁻¹ for 300 s. Mice were trained in three trials per day with a 30 min inter-trial interval for four consecutive days (12 trials total).

Open Field

The dimensions of the square open field arena (in cm) were 60 l × 60 w × 30 h. Animals were placed in the middle of the arena at the start of the test and remained in the arena for 30 min during which time their position was tracked using an overhead camera and analyzed using software (Limelight; Actimetrics).

Digging behavior

Mice were placed in a novel cage containing wood chip bedding (3-cm deep) in a brightly lit, open area. Animals remained in the testing cage for 15 min. Mouse digging time was manually scored using a stopwatch.

Voluntary Wheel Running

Mice of either sex were single-housed with wireless running wheels (MedAssociates) for at least 7 days. Total distance run was quantified for each day.

Virus Injection

For mice that received a viral injection (pAAV.Syn.Flex.GCaMP6f.WPRE.SV40, 100833-AAV9 from Addgene) prior to GRIN lens implant, we anesthetized mice with isoflurane (2% in O₂) and stereotactically injected virus at a rate of 250 nl min⁻¹ into the DLS using a microsyringe with a 28-gauge beveled-tip needle (Hamilton Company, Reno, NV). Stock titer virus was diluted 1:10 with sterile PBS. The coordinates for injection were: A/P +/- 0.05 mm, M/L +/- 1.8 mm, D/V -2.4mm (from dura). For all DV coordinates, we went 0.1 mm past the injection target and then withdrew the

syringe back to the target for the injection. After each injection, we left the syringe in place for 10 minutes and then slowly withdrew the syringe. We then sutured the scalp, injected analgesic (Buprenorphine SR, 1 mg kg⁻¹), as well as meloxicam (2x / day for 48 hours, 5 mg·kg⁻¹), and allowed the mice to recover for at least 14 days.

Silk fibroin coating of GRIN lenses

For animals in which we implanted a GRIN lens coated with a fibroin:AAV mixture, we autoclaved both GRIN lenses, and a small aliquot of 5% W/V silk fibroin solution, prior to coating. Then, the sterile fibroin solution was mixed 1:1 with stock titer AAV (pAAV.Syn.Flex.GCaMP6f.WPRE.SV40, 100833-AAV9 from Addgene). A microsyringe with a 23 gauge flat tip needle (Hamilton Company, Reno, NV) with a manual stereotaxic injector (Stoelting) was used to apply a 1μL droplet of the mixture to the tip of the GRIN lens in a biological safety cabinet. The mixture was allowed to dry slightly for 5 minutes with the droplet in contact with both the GRIN lens surface and the syringe needle tip before the syringe needle was removed, this prevented the mixture from coating the side of the lens. Once the needle was removed the droplet was allowed to fully dry to the surface of the lens for at least 1 hour before implanting. Lenses were stored at 4°C for implanting either the same day or within approximately 24 hours.

Surgical implantation of GRIN lens and viral transduction

Mice were anesthetized with isoflurane in O₂ (induction 4%; maintenance 2–3%). Following verification of anesthesia, hair was trimmed from the incision site, the scalp cleaned with three alternating swabs of betadine and 70% ethanol, and an incision made through the scalp to uncover the skull. The skull surface was cleaned and verified to be level by measuring the coordinates of bregma, lambda, and two sites +/- 2mm lateral to the midline. The M/L and A/P coordinates for implantation of the GRIN lens were marked on the skull (A/P +/- 0.05 mm, M/L +/- 2 mm) and a Dremel with a sterilized 0.6-mm tip was used to remove an approximately 2-mm circular region around the implant site. The skull was cleaned, dried, and scored crosswise with a scalpel.

While continuously perfusing the tissue with ACSF, the dura and cortex were removed using a 26 GA blunt-end needle connected to a vacuum line until the fibers of the corpus callosum became visible, at which point a 30 GA blunt-end needle was used to carefully remove the fibers of the callosum until striatal tissue became visible under a dissecting microscope. A 1-mm or 1.8-mm diameter GRIN lens was then lowered into the aspiration site (D/V -2.4 mm), held in place using a pipette tip and vacuum line attached to the stereotax. The lens was secured to the skull using dental cement (Metabond) and cyanoacrylate glue. Buprenorphine SR, (1 mg·kg⁻¹) was administered following surgery, as well as meloxicam (2x / day for 48 hours, 5 mg·kg⁻¹), dexamethasone (1x / day, 0.2 mg·kg⁻¹ for 7 days), and Sulfatrim (for 7 days in the drinking water). Mice were allowed to recover for 2–4 weeks before mounting the miniature microscope.

Miniscope Mounting and Imaging

Two-to-four weeks following surgery to implant the GRIN lens, the mouse was again anesthetized under the stereotax and a miniscope (UCLA V3 or V4) with a baseplate attached was lowered to the top focal plane of the GRIN lens. Once the optimal imaging plane was found, the baseplate was secured with dental cement (Metabond, Parkell).

In vivo Ca²⁺ imaging and Analysis

All mice were habituated to the weight of the miniscope over a 60-minute session in the open field. Following habituation, we placed mice in the open field for 60 minutes each day for 1–3 days with the miniscope attached. For analysis of calcium dynamics in dSPN cKO mice, we analyzed data from the first 20 minutes of day 1 of these sessions. In the majority of animals used in these

experiments, we used v3 miniscopes which required higher LED intensities compared to v4 miniscopes. Therefore, we limited recording to 20 minutes to limit photobleaching. For experiments testing fenobam, JZL and JNJ, 60 minutes of open field activity were recorded following IP injection. For these experiments we used v4 scopes which allowed for the use of lower LED intensities.

Fenobam (26 mg/kg) and JZL (10 mg/kg) were administered immediately prior to the test, following a 15-minute habituation. Due to its pharmacokinetics, JNJ was administered 30 minutes prior to beginning the test. Fenobam was diluted in 40% beta-cyclodextrin, 10% DMSO, JZL and JNJ were diluted in 10% DMSO and 90% corn oil.

We used open source V3 and V4 miniscopes (<https://github.com/Aharoni-Lab/Miniscope-v4>) that were connected to a coaxial cable which connected to a miniscope data acquisition board (DAQ) 3.3. The DAQ was connected to a computer via a USB 3.0 cable. Data was collected via the Miniscope QT Software (<https://github.com/Aharoni-Lab/Miniscope-DAQ-QT-Software>) at 20 frames per second. Simultaneous video recordings of behavior were performed at approximately 30 fps, and data was aligned posthoc using the time stamps collected by the miniscope software for each frame. Miniscopes and DAQ boards were purchased from Open Ephys, LabMaker, or parts were machined by Shylo Steitler (UCLA) and assembled.

Pre-processing and motion correction

Videos were preprocessed to remove sensor noise using a 2D FFT spatial frequency filter and lowpass butterworth filter (<https://github.com/Aharoni-Lab/Miniscope-v4/tree/master/Miniscope-v4-Denoising-Notebook>) and then motion correction was performed using the normcorr algorithm in the CaImAn package⁴³.

Identification of neurons (CNMF-E)

Cells were automatically identified using the CNMF-E algorithm⁴⁴. Recordings were grouped in batches of 1-5000 frames for processing, and spatially downsampled 2x during running of the CNMF-E algorithm. The maximum neuron diameter was determined based on manual observation of each of the motion corrected videos in ImageJ and the minimum local correlation value (Cn) and signal to noise ratio (SNR) for a pixel to be used as a cell “seed” was determined by visualizing the correlation image for a subset of recording frames of each video. The threshold for merging neurons was set at 0.65, neurons were merged if their distances were smaller than two pixels and they had highly correlated spatial shapes (correlation >0.8) and small temporal correlations (correlation <0.4). The output of the CNMF-E analysis performed on the video was then manually checked to ensure accuracy.

Analysis of Ca²⁺ activity

The cell Ca²⁺ activity traces from CNMF-E were processed first by scaling to the 99-percentile value of each cell’s activity trace during the recording session. The resulting normalized trace was then expressed as a Z-score by subtracting each sample from the mean and dividing by the standard deviation of the entire trace. Finally, this trace was downsampled to 5Hz (using the maximum value within the down sampled regions) and this result was used for further analysis.

Ca²⁺ event rate and correlation to velocity

A threshold crossing algorithm was used to detect peaks that were ≥ 2.5 s.d. above the baseline signal value for 1 second and the time of each event was calculated as the midpoint between the maximum value of the event and the preceding minimum¹¹. We used EZ track⁴⁵ to compute

the frame by frame velocity of the animal in the openfield arena. Using custom Python scripts we applied a 1 second median filter to this velocity trace and downsampled to 5Hz for alignment to the Ca^{2+} activity data.

Pairwise cell co-activity

The Jaccard index of pairwise coactivity was calculated following 1 second “forward-smoothing” of the Ca^{2+} activity traces as described previously ¹¹. The mean value of the Jaccard index was calculated as a function of the Euclidean distance between cells in a neuron pair and the mean values were binned into either 50 or 250 μm bins. These values were normalized to shuffled data sets where the Ca^{2+} traces for each cell were shuffled in time, using 1,000 independent shuffles of the binarized Ca^{2+} event traces for each cell. This data was used for statistical comparison of coactivity between groups as a function of distance, and behavioral state, i.e. moving or resting.

Spatial coordination index

To measure spatial coordination, the extent to which co-activity is greater among more proximal neurons versus more distal cells, during each frame of the video we calculated a previously used metric, the spatial coordination index (SCI) ¹¹. For each frame of the down-sampled video, we shuffled the pairwise distances for each neuron pair (1000 independent shuffles for each frame of the video), but not each cell’s Ca^{2+} activity signal. From these 1000 different distributions of distances between cell pairs, we calculated the cumulative distribution of active cell distances for the actual dataset and each shuffled dataset. We then performed a pair of Kolmogorov–Smirnov tests to determine if the cumulative distribution of active cell distances in each frame was smaller (i.e. left shifted or more “clustered”) or greater (right-shifted or less “clustered”) than that of the shuffled data. We took the logarithm (base ten) of the smaller p value from either test and used this value as the spatial coordination index. If a frame did not have any active cell pairs the index is 0. This analysis allows for visualization of how spatiotemporal dynamics change more specifically during movement onset.

Detection of movement onset

To most precisely detect instances of movement onset, specifically forward ambulatory motion, for our SCI analysis, we used the frame by frame velocity calculated from the mouse’s centroid location detected by ezTrack. We then filtered the velocity traces by determining if the animal’s centroid moved by more than 1cm from its location in the current frame at any point over the next 1 second of the recording. If not, we set the velocity to 0 for that frame. We then smoothed the resulting trace using a .167ms moving average. To find periods of movement onset that would allow us to more clearly measure whether there were consistent changes in spatial coordination, we identified where this velocity trace exceeded 0.5cm/sec for at least 2 seconds preceded by a period of rest for 4 seconds. We used the timestamps recorded by the miniscope software to find the matching period in our SCI signal. This allowed us to clearly measure how the degree of spatial coordination among dSPNs changes between rest and movement.

Slice procedure and in vitro Electrophysiological Recording

Parasagittal slices containing the dorsal striatum were prepared using standard techniques. Mice were deeply anesthetized (xylazine 10mg/kg and ketamine 100mg/kg i.p.) before undergoing transcardial perfusion with an ice-cold sucrose artificial cerebrospinal fluid (ACSF) solution containing (in mM): 85 NaCl, 2.5 KCl, 1.25 NaH_2PO_4 , 25 NaHCO_3 , 25 glucose, 75 sucrose, 0.5 CaCl_2 and 4 MgCl_2 , equilibrated with 95% O_2 and 5% CO_2 . The brain was removed and mounted on the stage of vibratome (Leica Microsystems, Inc). 350 μm thick sections were made in ice-cold sucrose ACSF. Slices were transferred to a recovery chamber containing the sucrose slicing ACSF solution, which was gradually exchanged for a normal ACSF containing (in mM): 125 NaCl, 2.4 KCl, 1.2 Na_2PO_4 , 25 NaHCO_3 , 25 glucose, 1 CaCl_2 and 2 MgCl_2 , while the slices were maintained at 30°C.

Individual slices were transferred to a recording chamber and visualized under DIC. Micro-Manager open source software was used for collecting the CCD camera image ⁴⁶[\[46\]](#). For extracellular recordings, slices were perfused with normal ACSF containing (in mM): 125 NaCl, 2.4 KCl, 1.2 Na₂PO₄, 25 NaHCO₃, 25 glucose, 2 CaCl₂ and 1 MgCl₂. Recording electrodes were manufactured from borosilicate glass pipettes and had resistances of 3-5 MΩ when filled with regular ACSF. For whole-cell voltage clamp experiments, recording electrodes were filled with internal solution containing (in mM) 95 CsF, 25 CsCl, 10 Cs-HEPES, 10 Cs-EGTA, 2 NaCl, 2 Mg-ATP, 10 QX-314, 5 TEA-Cl, 5 4-AP for recording of EPSCs, or 75 CsCH₃SO₃, 60 CsCl, 1 MgCl₂, 0.2 EGTA, 10 HEPES, 2 Mg-ATP 0.3 GTP-Na₂, 10 Na₂-phosphocreatine, 10 TEA, 5 QX-314. Striatal projection neurons (SPNs) were visually identified and were voltage clamped at -80mV. Miniature events (mEPSCs) were recorded in the presence of the Na⁺ channel blocker TTX (1μM) and bicuculline (10 μM). Analysis of mEPSCs was performed using Eventer software (<https://eventer.neuro.netlify.app/> [\[47\]](#)).

Histology

Following in vivo imaging experiments, mice were anesthetized using ketamine/xylazine and perfused with PBS containing 0.02% sodium nitrite and 4 mM MgSO₄ followed by 2% paraformaldehyde in 0.1 M sodium acetate buffer (pH 6.5) for 6 min, and 2% paraformaldehyde in 0.1 M sodium borate buffer (pH 8.5) for 12–18 min. The brains were removed from the skull, post-fixed overnight in 2% paraformaldehyde in 0.1 M sodium borate buffer (pH 8.5), then sectioned coronally at 50 μm on a Leica Vibratome VT1000s in PBS. The sections were collected and stored at 4 °C in PBS containing 0.02% sodium azide. Free-floating sections were rinsed for 30 min in PBS followed by four 15-min washes in PBS containing 0.1% Triton X-100 (Sigma, T8787) and 0.1% bovine serum albumin (BSA; Sigma, A4503). The sections were then blocked with 3% normal donkey serum (NDS; Jackson ImmunoResearch, 017-000-121) for 1 h and then incubated in primary antibody overnight (Mouse anti-GFP, Millipore, MAB3580).

Sections were washed three times for 15 min and then incubated in secondary antibody for 1 h and washed three times prior to mounting and imaging with a 20× objective on a Nikon A1 confocal microscope. Z-series were taken with a 0.82 μm step size.

Statistics

For comparisons of single mean values per mouse, for instance mean velocity per animal, we used non-parametric Mann Whitney U test or the Wilcoxon Signed Rank test. Where the same measurement was made multiple times, we used a repeated measures ANOVA. For cases where multiple different but correlated, non-independent, comparisons were made within mice, for instance in comparing mean event rates at different speeds or mean coactivity across neuron pairs at difference distances, we used a linear mixed effects model (LMM) to account for the non-independence of measurements made from the same animal. The model was implemented using the statsmodels python package (<https://www.statsmodels.org> [\[48\]](#)), the drug treatment or genotype was the independent variable, mean values measured per mouse are the dependent variables and individual mice are the group term. Where this model was used, we tested for the effect of genotype or drug treatment, and reported the significance for that measurement. In all figures, individual data points represent the mean of a group, and errors bars the standard error of the mean. N numbers for each group are specified in the figure legend.

Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. Code and summary data needed to replicate figures in the paper will be made publicly available in an archival database prior to the date of publication.

Figure 2 supplementary data 1

a) Averaged Ca^{2+} transients from vehicle and fenobam treated mice **b)** Fenobam treatment does not affect the distribution of event amplitudes (LMM, $p=0.246$, $z=-1.160$, $\beta=-0.555$, $\text{SE}=0.478$, 95% CI: $[-1.493\ 0.383]$, $N=8$ mice) **c)** Fenobam treatment does not affect the distribution of event lengths (LMM, $p=0.499$, $z=-0.676$, $\beta=-0.044$, $\text{SE}=0.065$, 95% CI: $[-0.173\ 0.084]$, $N=8$ mice).

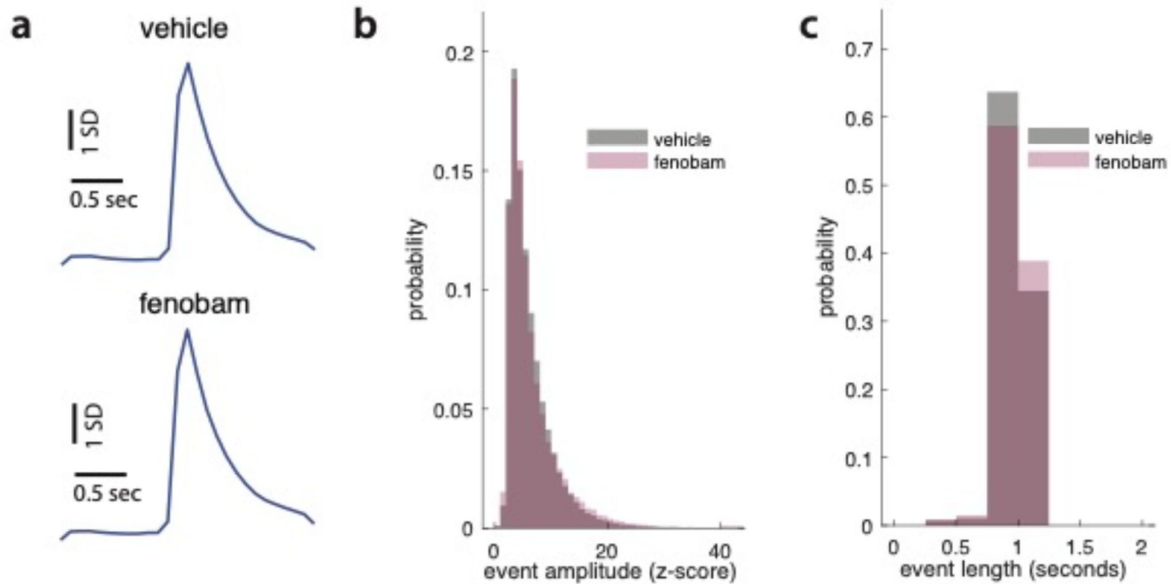


Figure 3 supplementary data.

a) Averaged Ca^{2+} transients from vehicle and JNJ treated mice **b)** JNJ treatment does not affect the distribution of event amplitudes (LMM, $p=0.955$, $z=0.056$, $\beta=0.048$, $\text{SE}=0.863$, 95% CI: $[-1.644\ 1.741]$, $N=5$ mice) **c)** JNJ treatment does not affect the distribution of event lengths (LMM, $p=0.913$, $z=0.110$, $\beta=0.021$, $\text{SE}=0.188$, 95% CI: $[-0.348\ 0.389]$, $N=5$ mice)

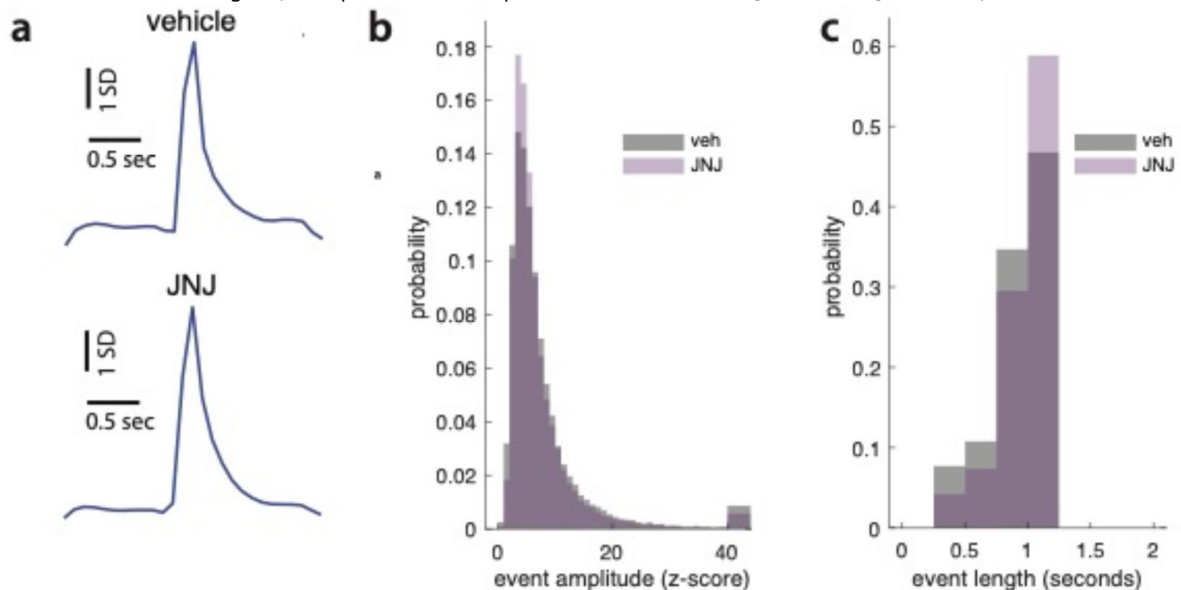


Figure 5 supplementary data 1.

a) Averaged Ca^{2+} transients from wt and cKO mice **b)** Wt and cKO mice show similar distributions of event amplitudes (LMM, $p=0.497$, $z=-0.679$, $\beta=-0.177$, $\text{SE}=0.261$, 95% CI: $[-0.690\ 0.335]$, $N=5$ mice) **c)** Wt and cKO mice show similar distributions of event length (LMM, $p=0.284$, $z=1.072$, $\beta=0.140$, $\text{SE}=0.130$, 95% CI: $[-0.116\ 0.395]$, $N=5$ mice)

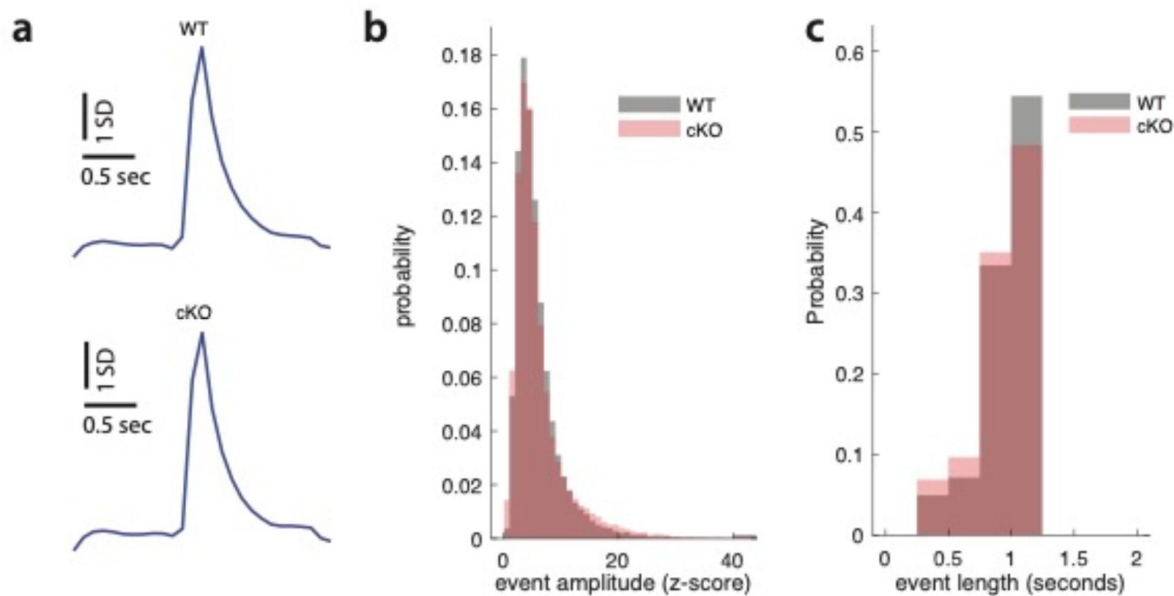


Figure 1 – supplementary data 1. Table 1.

Number of cells per animal and recording sessions from animals used in figure 1 [analysis](#).

Mouse	Total Cells	Sessions
19.2	318	5
1.2	78	2
20.2	311	2
24.2	20	2
25.2	307	4
26.2	176	2
2.7	136	3

Vehicle sessions for fenobam experiments

Mouse	Cells	Sessions
2.7	96	2
2.8	40	2
2.14	58	2
m974	293	1
m973	277	1
f979	164	1
f977	136	1
f976	387	1

Fenobam sessions

Mouse	Cells	Sessions
2.7	100	2
2.8	41	2
2.14	84	1
m974	91	1
m973	191	1
f979	29	1
f977	157	1
f976	219	1

Figure 2 – supplementary data 2. Table 2.

Number of cells per animal and recording sessions from animals used in figure 2 [analysis](#)

Vehicle Sessions for JNJ experiments

Mouse	Total Cells	Sessions
2.20	111	2
2.21	150	2
2.25	671	3
2.24	107	2
2.26	199	2

JNJ sessions

Mouse	Total Cells	Sessions
2.20	152	2
2.21	94	2
2.25	822	3
2.24	58	2
2.26	36	2

Vehicle Sessions for JZL experiments

Mouse	Total Cells	Sessions
2.8	150	4
2.14	129	2
2.19	203	2
2.20	80	2
2.26	22	2

JZL sessions

Mouse	Total Cells	Sessions
2.8	65	2
2.14	33	2
2.19	29	1
2.20	120	2
2.26	20	2

Figure 3 – supplementary data 2. Table 3.

Vehicle Sessions for JNJ experiments. Number of cells per animal and recording sessions from animals used in figure 3 [analysis](#)

WT (D1 cre ; *GRM5* +/+) – dSPNs

Mouse	Cells	Sessions
2.1	23	1
2.19	83	2
1.13	35	1
1.26	22	2
1.27	45	1
1.32	237	2
1.33	128	2
1.35	280	3
1.39	159	3

cKO (D1 cre ; *GRM5* f/f) - dSPNs

Mouse	Cells	Sessions
1.9	143	1
1.11	202	1
1.12	139	2
1.18	104	2
1.38	377	3
1.41	2	1

WT (D1 cre ; *GRM5* +/+) – iSPNs

Mouse	Cells	Sessions
DO_1	114	1
DO_2	125	1
DO_3	39	1

cKO (D1 cre ; *GRM5* f/f) – iSPNs

Mouse	Cells	Sessions
DO_4	63	1
DO_5	116	1
DO_6	221	1

Figure 5 supplementary data 2. Table 5.

WT (D1 cre; *GRM5* +/+) – dSPNs. Number of cells per animal and recording sessions from animals in figure 5 [↗](#).

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Author contributions

Conceptualization: JJM, JX, JGP, AC Methodology: JJM, JX, JGP, AC Investigation: JJM, JX, NHY, SY, TN, JA Visualization: JJM, JGP, AC Supervision: JJM, JGP, AC Writing—original draft: JJM, AC Writing—review & editing: JJM, AC, JGP

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References

1. Klaus A., Alves da Silva J., Costa R.M. (2019) **What, If, and When to Move: Basal Ganglia Circuits and Self-Paced Action Initiation** *Annu Rev Neurosci* **42**:459–483 <https://doi.org/10.1146/annurev-neuro-072116-031033> | Google Scholar
2. Wolff S.B.E., Ko R., Olveczky B.P. (2022) **Distinct roles for motor cortical and thalamic inputs to striatum during motor skill learning and execution** *Sci Adv* **8**:eabk0231 <https://doi.org/10.1126/sciadv.abk0231> | Google Scholar
3. Dang M.T., Yokoi F., Yin H.H., Lovinger D.M., Wang Y., Li Y. (2006) **Disrupted motor learning and long-term synaptic plasticity in mice lacking NMDAR1 in the striatum** *Proc Natl Acad Sci U S A* **103**:15254–15259 <https://doi.org/10.1073/pnas.0601758103> | Google Scholar
4. Rothwell P.E., Hayton S.J., Sun G.L., Fuccillo M.V., Lim B.K., Malenka R.C. (2015) **Input-and Output-Specific Regulation of Serial Order Performance by Corticostriatal Circuits** *Neuron* **88**:345–356 <https://doi.org/10.1016/j.neuron.2015.09.035> | Google Scholar
5. Yin H.H., Mulcare S.P., Hilario M.R., Clouse E., Holloway T., Davis M.I., Hansson A.C., Lovinger D.M., Costa R.M. (2009) **Dynamic reorganization of striatal circuits during the acquisition and consolidation of a skill** *Nat Neurosci* **12**:333–341 <https://doi.org/10.1038/nn.2261> | Google Scholar
6. Hwang F.J., Roth R.H., Wu Y.W., Sun Y., Kwon D.K., Liu Y., Ding J.B. (2022) **Motor learning selectively strengthens cortical and striatal synapses of motor engram neurons** *Neuron* **110**:2790–2801 <https://doi.org/10.1016/j.neuron.2022.06.006> | Google Scholar
7. Gerfen C.R., Surmeier D.J. (2011) **Modulation of striatal projection systems by dopamine** *Annu Rev Neurosci* **34**:441–466 <https://doi.org/10.1146/annurev-neuro-061010-113641> | Google Scholar
8. Nelson A.B., Kreitzer A.C. (2014) **Reassessing models of basal ganglia function and dysfunction** *Annu Rev Neurosci* **37**:117–135 <https://doi.org/10.1146/annurev-neuro-071013-013916> | Google Scholar
9. Cui G., Jun S.B., Jin X., Pham M.D., Vogel S.S., Lovinger D.M., Costa R.M. (2013) **Concurrent activation of striatal direct and indirect pathways during action initiation** *Nature* **494**:238–242 <https://doi.org/10.1038/nature11846> | Google Scholar
10. Barbera G., Liang B., Zhang L., Gerfen C.R., Culurciello E., Chen R., Li Y., Lin D.T. (2016) **Spatially Compact Neural Clusters in the Dorsal Striatum Encode Locomotion Relevant Information** *Neuron* **92**:202–213 <https://doi.org/10.1016/j.neuron.2016.08.037> | Google Scholar
11. Parker J.G., Marshall J.D., Ahanonu B., Wu Y.W., Kim T.H., Grewe B.F., Zhang Y., Li J.Z., Ding J.B., Ehlers M.D., Schnitzer M.J. (2018) **Diametric neural ensemble dynamics in parkinsonian and dyskinetic states** *Nature* **557**:177–182 <https://doi.org/10.1038/s41586-018-0090-6> | Google Scholar

12. Klaus A., Martins G.J., Paixao V.B., Zhou P., Paninski L., Costa R.M (2017) **The Spatiotemporal Organization of the Striatum Encodes Action Space** *Neuron* **95**:1171–1180 <https://doi.org/10.1016/j.neuron.2017.08.015> | Google Scholar
13. Yun S., Yang B., Anair J.D., Martin M.M., Fleps S.W., Pamukcu A., Yeh N.H., Contractor A., Kennedy A., Parker J.G (2023) **Antipsychotic drug efficacy correlates with the modulation of D1 rather than D2 receptor-expressing striatal projection neurons** *Nat Neurosci* **26**:1417–1428 <https://doi.org/10.1038/s41593-023-01390-9> | Google Scholar
14. Wilson C.J., Groves P.M (1981) **Spontaneous firing patterns of identified spiny neurons in the rat neostriatum** *Brain Res* **220**:67–80 [https://doi.org/10.1016/0006-8993\(81\)90211-0](https://doi.org/10.1016/0006-8993(81)90211-0) | Google Scholar
15. Wilson C.J., Kawaguchi Y (1996) **The origins of two-state spontaneous membrane potential fluctuations of neostriatal spiny neurons** *J Neurosci* **16**:2397–2410 <https://doi.org/10.1523/JNEUROSCI.16-07-02397.1996> | Google Scholar
16. Guo Q., Wang D., He X., Feng Q., Lin R., Xu F., Fu L., Luo M (2015) **Whole-brain mapping of inputs to projection neurons and cholinergic interneurons in the dorsal striatum** *PLoS One* **10**:e0123381 <https://doi.org/10.1371/journal.pone.0123381> | Google Scholar
17. Wall N.R., De La Parra M., Callaway E.M., Kreitzer A.C. (2013) **Differential innervation of direct-and indirect-pathway striatal projection neurons** *Neuron* **79**:347–360 <https://doi.org/10.1016/j.neuron.2013.05.014> | Google Scholar
18. Wu Y.W., Kim J.I., Tawfik V.L., Lalchandani R.R., Scherrer G., Ding J.B (2015) **Input-and cell-type-specific endocannabinoid-dependent LTD in the striatum** *Cell Rep* **10**:75–87 <https://doi.org/10.1016/j.celrep.2014.12.005> | Google Scholar
19. Li W., Pozzo-Miller L (2020) **Dysfunction of the corticostriatal pathway in autism spectrum disorders** *J Neurosci Res* **98**:2130–2147 <https://doi.org/10.1002/jnr.24560> | Google Scholar
20. Surmeier D.J., Plotkin J., Shen W (2009) **Dopamine and synaptic plasticity in dorsal striatal circuits controlling action selection** *Curr Opin Neurobiol* **19**:621–628 <https://doi.org/10.1016/j.conb.2009.10.003> | Google Scholar
21. Gerdeman G.L., Ronesi J., Lovinger D.M (2002) **Postsynaptic endocannabinoid release is critical to long-term depression in the striatum** *Nat Neurosci* **5**:446–451 <https://doi.org/10.1038/nn832> | Google Scholar
22. Bagetta V., Picconi B., Marinucci S., Sgobio C., Pendolino V., Ghiglieri V., Fusco F.R., Giampa C., Calabresi P (2011) **Dopamine-dependent long-term depression is expressed in striatal spiny neurons of both direct and indirect pathways: implications for Parkinson's disease** *J Neurosci* **31**:12513–12522 <https://doi.org/10.1523/JNEUROSCI.2236-11.2011> | Google Scholar
23. Kreitzer A.C., Malenka R.C (2007) **Endocannabinoid-mediated rescue of striatal LTD and motor deficits in Parkinson's disease models** *Nature* **445**:643–647 <https://doi.org/10.1038/nature05506> | Google Scholar
24. Marshall J.J., Xu J., Contractor A (2018) **Kainate Receptors Inhibit Glutamate Release Via Mobilization of Endocannabinoids in Striatal Direct Pathway Spiny Projection Neurons** *J Neurosci* **38**:3901–3910 <https://doi.org/10.1523/JNEUROSCI.1788-17.2018> | Google Scholar

25. Chevalleyre V., Takahashi K.A., Castillo P.E (2006) **Endocannabinoid-mediated synaptic plasticity in the CNS** *Annu Rev Neurosci* **29**:37–76 <https://doi.org/10.1146/annurev.neuro.29.051605.112834> | Google Scholar
26. Jackman S.L., Chen C.H., Chettih S.N., Neufeld S.Q., Drew I.R., Agba C.K., Flaquer I., Stefano A.N., Kennedy T.J., Belinsky J.E., et al. (2018) **Silk Fibroin Films Facilitate Single-Step Targeted Expression of Optogenetic Proteins** *Cell Rep* **22**:3351–3361 <https://doi.org/10.1016/j.celrep.2018.02.081> | Google Scholar
27. Aharoni D., Hoogland T.M (2019) **Circuit Investigations With Open-Source Miniaturized Microscopes: Past, Present and Future** *Front Cell Neurosci* **13** <https://doi.org/10.3389/fncel.2019.00141> | Google Scholar
28. Shin J.H., Song M., Paik S.B., Jung M.W (2020) **Spatial organization of functional clusters representing reward and movement information in the striatal direct and indirect pathways** *Proc Natl Acad Sci U S A* **117**:27004–27015 <https://doi.org/10.1073/pnas.2010361117> | Google Scholar
29. Uchigashima M., Narushima M., Fukaya M., Katona I., Kano M., Watanabe M (2007) **Subcellular arrangement of molecules for 2-arachidonoyl-glycerol-mediated retrograde signaling and its physiological contribution to synaptic modulation in the striatum** *J Neurosci* **27**:3663–3676 <https://doi.org/10.1523/JNEUROSCI.0448-07.2007> | Google Scholar
30. Xu J., Marshall J.J., Kraniotis S., Nomura T., Zhu Y., Contractor A (2021) **Genetic disruption of Grm5 causes complex alterations in motor activity, anxiety and social behaviors** *Behav Brain Res* **411** <https://doi.org/10.1016/j.bbr.2021.113378> | Google Scholar
31. Conde-Ceide S., Martinez-Vituro C.M., Alcazar J., Garcia-Barrantes P.M., Lavreysen H., Mackie C., Vinson P.N., Rook J.M., Bridges T.M., Daniels J.S., et al. (2015) **Discovery of VU0409551/JNJ-46778212: An mGlu5 Positive Allosteric Modulator Clinical Candidate Targeting Schizophrenia** *ACS Med Chem Lett* **6**:716–720 <https://doi.org/10.1021/acsmedchemlett.5b00181> | Google Scholar
32. Xu J., Zhu Y., Contractor A., Heinemann S.F (2009) **mGluR5 has a critical role in inhibitory learning** *J Neurosci* **29**:3676–3684 <https://doi.org/10.1523/JNEUROSCI.5716-08.2009> | Google Scholar
33. Gerfen C.R., Paletzki R., Heintz N (2013) **GENSAT BAC cre-recombinase driver lines to study the functional organization of cerebral cortical and basal ganglia circuits** *Neuron* **80**:1368–1383 <https://doi.org/10.1016/j.neuron.2013.10.016> | Google Scholar
34. Madisen L., Zwingman T.A., Sunkin S.M., Oh S.W., Zariwala H.A., Gu H., Ng L.L., Palmiter R.D., Hawrylycz M.J., Jones A.R., et al. (2010) **A robust and high-throughput Cre reporting and characterization system for the whole mouse brain** *Nat Neurosci* **13**:133–140 <https://doi.org/10.1038/nn.2467> | Google Scholar
35. Taverna S., Ilijic E., Surmeier D.J (2008) **Recurrent collateral connections of striatal medium spiny neurons are disrupted in models of Parkinson's disease** *J Neurosci* **28**:5504–5512 <https://doi.org/10.1523/JNEUROSCI.5493-07.2008> | Google Scholar

36. Saunders A., Johnson C.A., Sabatini B.L (2012) **Novel recombinant adeno-associated viruses for Cre activated and inactivated transgene expression in neurons** *Front Neural Circuits* **6** <https://doi.org/10.3389/fncir.2012.00047> | Google Scholar
37. McElvain L.E., Chen Y., Moore J.D., Brigidi G.S., Bloodgood B.L., Lim B.K., Costa R.M., Kleinfeld D (2021) **Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon** *Neuron* **109**:1721–1738 <https://doi.org/10.1016/j.neuron.2021.03.017> | Google Scholar
38. Parent A., De Bellefeuille L. (1982) **Organization of efferent projections from the internal segment of globus pallidus in primate as revealed by fluorescence retrograde labeling method** *Brain Res* **245**:201–213 [https://doi.org/10.1016/0006-8993\(82\)90802-2](https://doi.org/10.1016/0006-8993(82)90802-2) | Google Scholar
39. Shen W., Plotkin J.L., Francardo V., Ko W.K., Xie Z., Li Q., Fieblinger T., Wess J., Neubig R.R., Lindsley C.W., et al. (2015) **M4 Muscarinic Receptor Signaling Ameliorates Striatal Plasticity Deficits in Models of L-DOPA-Induced Dyskinesia** *Neuron* **88**:762–773 <https://doi.org/10.1016/j.neuron.2015.10.039> | Google Scholar
40. Barnes S.A., Pinto-Duarte A., Kappe A., Zembrzycki A., Metzler A., Mukamel E.A., Lucero J., Wang X., Sejnowski T.J., Markou A., Behrens M.M (2015) **Disruption of mGluR5 in parvalbumin-positive interneurons induces core features of neurodevelopmental disorders** *Mol Psychiatry* **20**:1161–1172 <https://doi.org/10.1038/mp.2015.113> | Google Scholar
41. Gong S., Doughty M., Harbaugh C.R., Cummins A., Hatten M.E., Heintz N., Gerfen C.R (2007) **Targeting Cre Recombinase to Specific Neuron Populations with Bacterial Artificial Chromosome Constructs** *The Journal of Neuroscience* **27**:9817–9823 | Google Scholar
42. Madisen L., Zwingman T.A., Sunkin S.M., Oh S.W., Zariwala H.A., Gu H., Ng L.L., Palmiter R.D., Hawrylycz M.J., Jones A.R., et al. (2010) **A robust and high-throughput Cre reporting and characterization system for the whole mouse brain** *Nature neuroscience* **13**:133–140 <https://doi.org/10.1038/nn.2467> | Google Scholar
43. Giovannucci A., Friedrich J., Gunn P., Kalfon J., Brown B.L., Koay S.A., Taxidis J., Najafi F., Gauthier J.L., Zhou P., et al. (2019) **CaImAn an open source tool for scalable calcium imaging data analysis** *eLife* **8** <https://doi.org/10.7554/eLife.38173> | Google Scholar
44. Zhou P., Resendez S.L., Rodriguez-Romaguera J., Jimenez J.C., Neufeld S.Q., Giovannucci A., Friedrich J., Pnevmatikakis E.A., Stuber G.D., Hen R., et al. (2018) **Efficient and accurate extraction of in vivo calcium signals from microendoscopic video data** *eLife* **7** <https://doi.org/10.7554/eLife.28728> | Google Scholar
45. Pennington Z.T., Diego K.S., Francisco T.R., LaBanca A.R., Lamsifer S.I., Liobimova O., Shuman T., Cai D.J (2021) **ezTrack-A Step-by-Step Guide to Behavior Tracking** *Curr Protoc* **1**:e255 <https://doi.org/10.1002/cpz1.255> | Google Scholar
46. Edelstein A.D., Tsuchida M.A., Amodaj N., Pinkard H., Vale R.D., Stuurman N (2014) **Advanced methods of microscope control using muManager software** *J Biol Methods* **1** <https://doi.org/10.14440/jbm.2014.36> | Google Scholar

Author information

John J Marshall

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States

ORCID iD: [0000-0002-6481-8413](https://orcid.org/0000-0002-6481-8413)

For correspondence: j-marshall@northwestern.edu

Jian Xu

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States

Nai-Hsing Yeh

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States

Seongsik Yun

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States

Toshihiro Nomura

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States

ORCID iD: [0000-0001-8786-831X](https://orcid.org/0000-0001-8786-831X)

John N Armstrong

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States

Jones G Parker

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States, Department of Psychiatry and Behavioral Sciences, Feinberg School of Medicine, Northwestern University, Chicago United States, Department of Pharmacology, Feinberg School of Medicine, Northwestern University, Chicago United States

ORCID iD: [0000-0001-9302-1545](https://orcid.org/0000-0001-9302-1545)

For correspondence: Jones.parker@northwestern.edu

Anis Contractor

Department of Neuroscience, Feinberg School of Medicine, Northwestern University, Chicago United States, Department of Psychiatry and Behavioral Sciences, Feinberg School of Medicine, Northwestern University, Chicago United States, Department of Neurobiology, Weinberg School of Arts and Sciences, Northwestern University, Evanston United States

ORCID iD: [0000-0002-5131-2536](https://orcid.org/0000-0002-5131-2536)

For correspondence: a-contractor@northwestern.edu

Editors

Reviewing Editor

Nicole Calakos

Duke Medical Center, Durham, United States of America

Senior Editor

John Huguenard

Stanford University School of Medicine, Stanford, United States of America

Joint Public Review:

Marshall et al describe the effects of altering metabotropic glutamate receptor 5 activity on activity of D1 receptor expressing spiny projection neurons in dorsolateral striatum focusing on two states - locomotion and rest. The authors examine effects of dSPN-specific constitutive mGlu5 deletion in several motor tests to arrive at this finding. Effects of inhibiting the degradation of the endocannabinoid 2-arachidonoyl glycerol are also examined. Overall, this is a valuable study that provides solid new information of relevance to movement disorders and possibly psychosis.

The combination of in vivo cellular calcium imaging, pharmacology, receptor knockout and movement analysis is effectively used. The main findings do not involve gross firing rates or numbers of active neurons, but rather are revealed by specialized measures involving Jaccard coefficient and an assessment of coactivity. The authors conclude that mGlu5 expressed in dSPNs contributes to movement through effects on clustered spatial coactivity of dSPNs. More specifically, reduced mGluR5 increases coactivity during rest (defined as low velocity periods) but not during locomotion periods. The authors observe a role for mGlu5 expression in dSPNs in modulating the frequency of mEPSCs, suggesting a role in presynaptic neurotransmitter release. Some data suggesting the story may be different in the other major SPN subpopulation (iSPNs) are also presented but these studies are relatively underdeveloped leaving some ambiguity as to how cell-selective the findings are. In addition, an occlusion experiment in which the pharmacological mGluR5 agents are delivered to the dSPN mGluR5 KO to clarify if other sites of action are involved beyond the proposed D1-expressing neurons is missing. Finally, the authors present a working model that sets the stage for future experimentation. Overall, this study provides an important and detailed assessment of mGluR5 contributions to striatal circuit function and behavior.

Remaining concerns include:

(1) To clarify that dSPNs are sole site of action, it is necessary to examine effects of the mGlu5 NAM in the dSPN mGlu5 cKO mice. If the effects of the two manipulations occluded one another this would certainly support the hypothesis that the drug effects are mediated by receptors expressed in dSPNs. A similar argument can be made for examining effects of the JNJ PAM in the cKO mice.

(2) There is a concern that the D1 Cre line used (Ey262), which may also target cortical neurons expands the interpretation of the study beyond the striatal populations. Further discussion of this point, particularly in the interpretation of the mGluR5 cKO experiments, would provide a better understanding of the contribution of the paper.

(3) The use of CsF-based whole-cell internal solutions has caused concern in some past studies due to possible interference with G-protein, phosphatase and channel function (<https://www.sciencedirect.com/science/article/abs/pii/S1044743104000296>, <https://www.jneurosci.org/content/jneuro/6/10/2915.full.pdf>). It is reassuring the DHPG-induced LTD was still observable with this solution. However, it might be worth examining this plasticity with a

different internal to ensure that the magnitude of the agonist effect is not altered by this manipulation.

(4) Behavioral resolution of actions at low velocity that are termed "rest" are not explored in this study. Thus, a remaining ambiguity is whether the activities in rest include only periods of immobility or other low-velocity activities such as grooming or rearing.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

"Can the authors offer a hypothesis as to how decreased coactivity promotes increased movement velocity."

In our revision we have added an additional metric measuring how spatial coactivity changes during movement onset, the spatial correlation index, which replicates a previous finding that co-activity among proximal neurons is statistically greater surrounding movement onset. We did not find, as outlined in the revision, that mGluR5 manipulations significantly altered this relationship. Our data therefore shows, consistent with that shown previously, that ensembles of dSPNs that are co-active during movement onset, in particular ambulatory movement, are more likely to contain neurons that are closer together and the neurons are highly active. In contrast, rest ensembles contain neurons that are less active but have more highly correlated activity, across all pairwise distances. Additionally, mGluR5 inhibition, genetic or pharmacological, promotes the activation of rest ensembles but does not affect the properties of movement ensembles. Previous studies (e.g. Klaus A. et al., 2017) have shown that neurons in rest ensembles are, in general, unlikely to also be members of movement ensembles. We therefore hypothesize that corticostriatal synapses onto SPNs of rest ensembles are more likely, during spontaneous behavior, to have reduced synaptic weight due to mGluR5 signaling, potentially due to eCB mediated inhibition of neurotransmitter release. Therefore, when we inhibit mGluR5 at these synapses, we increase synaptic weight and increase the probability of activation of this coordinated rest ensemble, which suppresses movement. If, on the other hand, the synapses that govern activation of neurons in movement ensembles have a higher weight, they may be unaffected by mGluR5 inhibition.

The use of the Jaccard similarity index in this study is not intuitive and not fully explained by the methods or the diagram in Figure 1.

We have added more detail to the paper to explain the methodology of the jaccard similarity measure. The advantage of this method is that it specifically captures cells that are jointly active, as opposed to jointly inactive and is therefore useful for capturing co-activity in our sparsely active Ca^{2+} imaging data.

The analysis of a possible 2-AG role in the mGlu5 mediated processes is incomplete.

We agree that, as an experiment to outline which endocannabinoids are involved in modulating synaptic strength through mGluR5, this experiment alone is not sufficient.

However, our main focus in this paper is how manipulations of mGluR5 affect the spatiotemporal dynamics of dSPNs and we chose not to focus on specific mechanisms of endocannabinoid signaling, though these would certainly be interesting to investigate further in vivo.

It would seem to be a simple experiment to examine effects of the mGlu5 NAM in the dSPN mGlu5 cKO mice. If effects of the two manipulations occluded one another this would certainly support the hypothesis that the drug effects are mediated by receptors expressed in dSPNs. A similar argument can be made for examining effects of the JNJ PAM in the cKO mice.

We agree that this experiment would be valuable and extend our findings presented in the paper, however, it has practically been outside the scope of the current work.

Reviewer #2 (Public review):

Pharmacological and genetic manipulations of mGluR5 do not differentially/preferentially modulate the activity of proximal vs distal dSPNs, therefore, it could also be interpreted that mGluR5 is blanketly boosting/suppressing all dSPN activity as opposed to differential proximal/distal spatial relationships.

As in the response to reviewer 1 above, we have added additional clarification to the text explaining that our manipulations do not differentially affect the co-activity of proximal vs distal dSPNs, this is also quantified throughout the text using the spatial coordination index. However, we disagree that “it could also be interpreted that mGluR5 is blanketly boosting/suppressing all dSPN activity” as we do not observe statistically significant changes in the event rate following either pharmacological or genetic manipulations of mGluR5. Rather, we consistently observe statistically significant changes in co-activity among neurons, the extent to which activity of active neurons during either rest or movement are correlated with each other. This is the central finding of our manuscript, inhibiting or potentiating mGluR5 signaling alters behavior, not by blanket suppression or enhancement of the activity as measured using the event rate, of dSPNs, but by affecting their ensemble dynamic properties. Co-activity during rest versus ambulatory movement is statistically greater in both proximal and distal cells and inhibiting mGluR5 increases this co-activity and decreases movement.

For these analyses of prox vs distal and all others, please include the detail of how many proximal vs distal cells were involved and per subject.

We have added a supplemental table that details the number of cells included per subject in all analyses

Ln. 151-152: Please provide data concerning how volumes of infectivity differ between injecting AAV vs. coating the lens? If these numbers are very different, this could impact the number of Jaccard pairings and bias results.

While viral injection may lead to a larger volume of expression, with this one photon imaging method only those cells within ~200 microns of the edge of the lens will be able to be resolved, therefore practically, if there is an additional volume of infected tissue outside of the field of view of the lens, it would not affect the results as these neurons will not be resolved by the endoscope camera. Accordingly, the average number of cells detected per session is very similar following each approach (mean # of cells per session with coating 90.93 ± 23.69 cells, with viral injection 90.03 ± 29.29 cells)

Is mGluR5 affecting dSPN activity in other measures beyond co-activity and rate? Does the amplitude of events change?

We have added supplemental data for figures 2, 3, and 5 demonstrating that manipulations of mGluR5 do not affect the amplitude or length of Ca^{2+} events included in the analysis.

What is the model of mGluR5 signaling in a resting state vs. movement? What other behaviors are occurring when the mouse is in a low velocity "resting state" (0-0.5 cm/s). If this includes other forms of movement (i.e. rearing, grooming) then the animal really isn't in a resting state. This is not mentioned in the open field behavior section of the methods and should be described (Ln. 486) in addition to greater explanation of what behavior measures were obtained from the video tracking software (only locomotion?)

It would be very interesting to determine if during "rest," when the animals is not engaged in ambulatory behavior, it may be engaged in some fine motor behavior. However, the resolution of the cameras used to measure locomotor activity in this dataset does not allow us to do this.

There is large variability in co-activity in proximal dSPNs when animals are "resting" (2j). Could this be explained by different behavior states within your definition of "rest"?

We agree that if the animal is engaging in fine motor behavior that we cannot resolve with our behavior setup, this could produce some variability in coactivity. However, as shown previously (e.g. Klaus A. et al., 2017), ensembles active when the animal is not moving (our definition of "resting"), regardless of additional fine motor behaviors the animal may be engaged in when not moving, are substantially different than those ensembles that are active when the animal is moving. We therefore expect that this may limit, although potentially not eliminate, variability due to different behavioral states we may have grouped into our "resting" category. Unfortunately, as mentioned above, we are not able to resolve variations in fine motor output in this behavioral data.

Have you performed IHC, ISH or another measure to validate D1 cell specific cKO?

The mGluR5^{loxP/loxP} mice used in this study were characterized previously by our lab (Xu et al., 2009), we used the same mice here with a different, but also published and characterized Cre-driver line, Drd1a-Cre Ey262 (Gerfen et al., 2013).

Why are the "Mean Norm Co-activity" values in 5e so high in this experiment relative to figures 2-4?

In experiments where we treated the same animal with vehicle and a drug (i.e., experiments in Figure 2 and 3), we normalized the values for each animal in the drug treatment group to the distal bin of that animal following vehicle treatment. This allowed us to more clearly resolve the changes within each animal due to drug treatment. As comparisons in the data in figure 5 d-f are between different animals (rather than different treatments of the same animal) we could not perform this normalization procedure.

Reviewer #3 (Public review):

Some D1 Cre lines have expression in the cortex. Which specific Cre line was used in this study?

We used, Drd1a-Cre Ey262. This is included in methods.

The text says JNJ treatment increased locomotor speed (Figure 3b) and increased the duration but not frequency of movement bouts (Figure 3c, d). However, the statistics of the figure legends say: however the change in mean velocity (3b) is not significant

($p=0.060$, $U=3$, Mann-Whitney U test), nor is the mean bout length during vehicle and JNJ ($p=0.060$, $U=3$, Mann-Whitney U test) (3d) Comparison of mean number of bouts of each animal during vehicle and JNJ ($p=0.403$, $U=8$, Mann-Whitney U test).

This has been corrected to indicate only the change in time spend at rest is statistically significant.

This effect was most pronounced during periods of rest (Figure 3i, j). The decrease was only in rest? Are the colors in Figure 3J inverted? Therefore, JNJ treatment had effects that were qualitatively the inverse to the effects of fenobam on locomotion and dSPN activity.

We have corrected the text to state that, overall, and during periods of rest but not movement, JNJ had effects that were qualitatively the opposite of fenobam.

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